

Computational Knot Theory

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INTRODUCTION & DEFINITIONS

INTRODUCTION

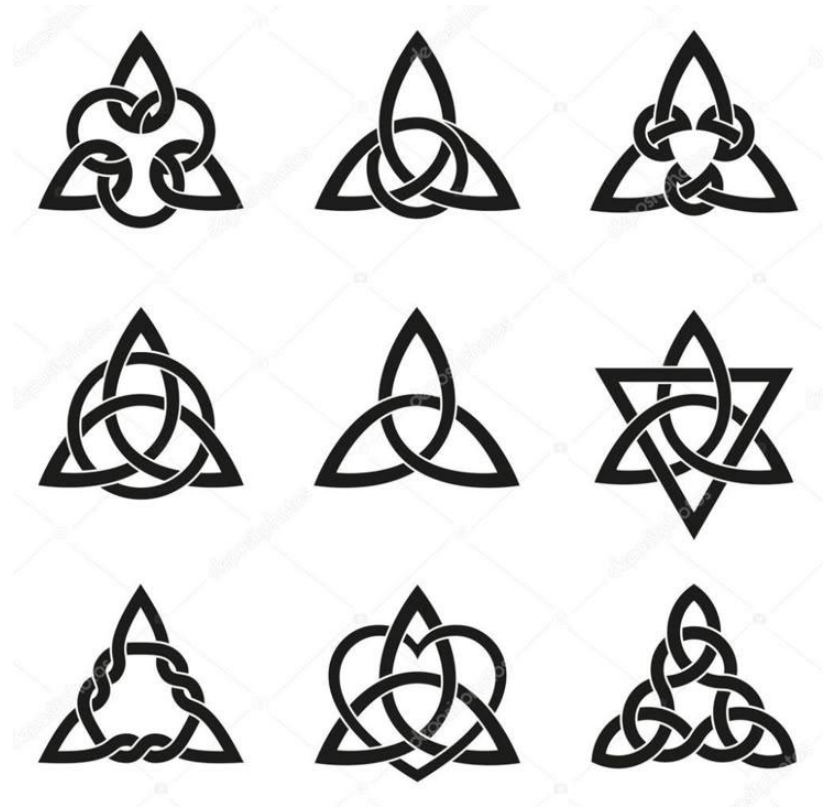
WHAT IS A KNOT ?

Informally, a knot may be thought of as a closed piece of string embedded in three dimensional space such that the string does not intersect itself. The ends of the string are joined together to form a closed loop.

Formally, a knot is defined as a homeomorphic image of a circle in three-dimensional Euclidean space i.e

Each knot K equals $h(S_1)$

for some homeomorphism $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$



Knots in celtic artwork

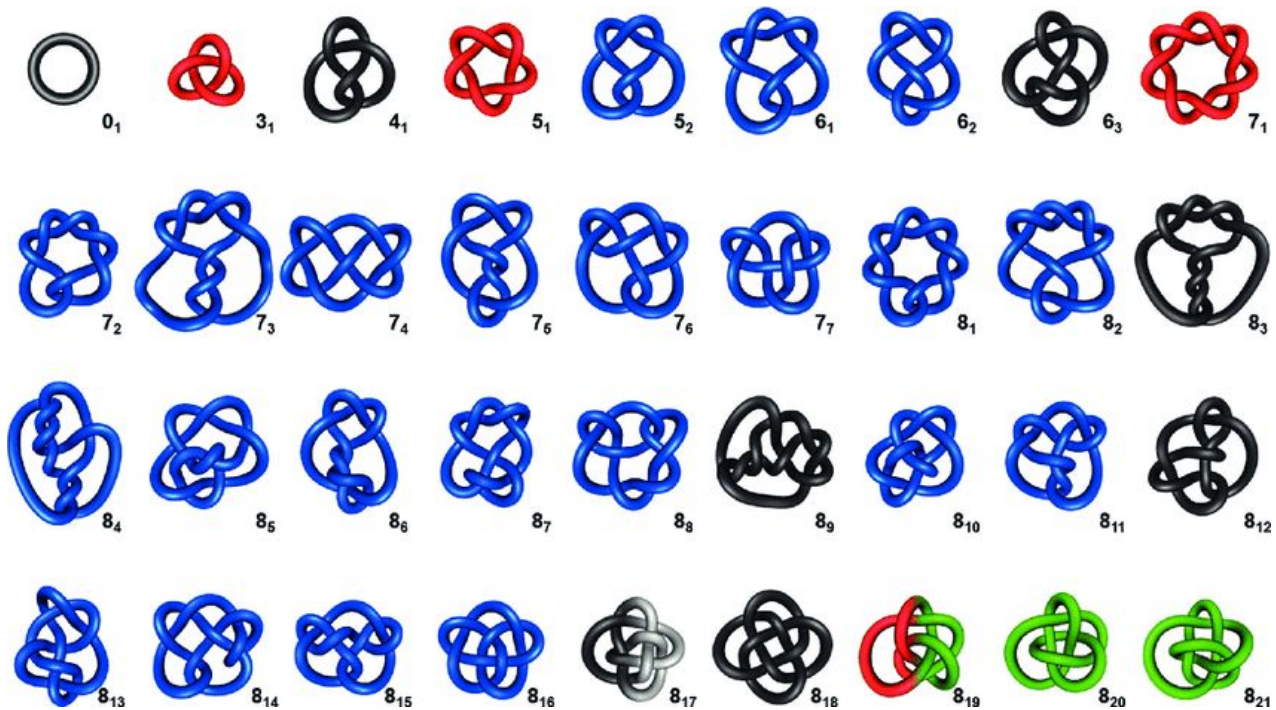
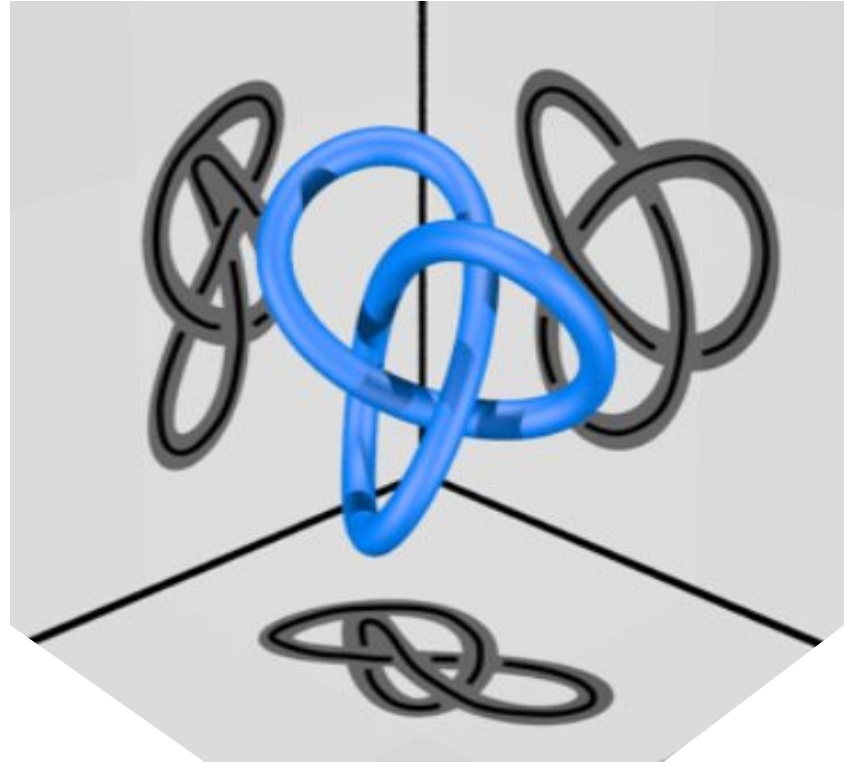


Table of knots with crossing number less than equal to 8

KNOT PROJECTIONS AND DIAGRAMS

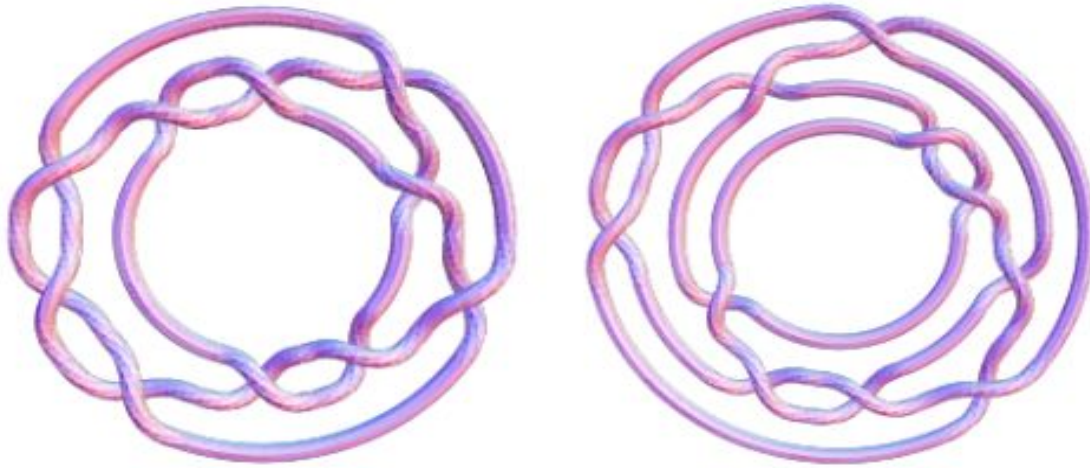
To understand and study knots, we project the knot onto a plane. The projections we are concerned with are called regular projection where not more than two points are projected onto the same point. At each crossing, we show which strand passes over and which passes under, by representing the under-strand by a broken line for each crossing. This is called diagram of the knot.

If two knots have the same regular projection then they are equivalent.



EQUIVALENCE OF KNOTS

Two knots K_1 and K_2 are said to be equivalent if there exists a homeomorphism $h: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $h(K_1) = K_2$. This clearly defines an equivalence relation on the set of knots. A knot that is equivalent to the standard unit circle is called a trivial knot (or unknot).



Perko Pair

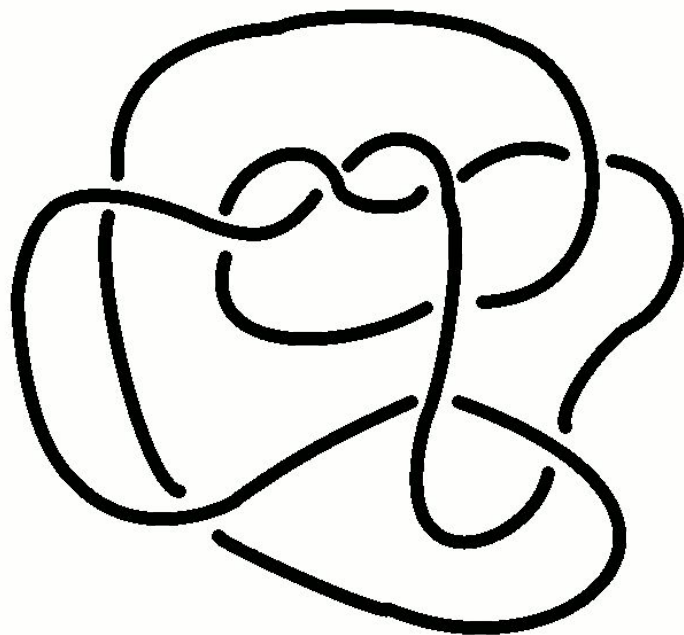
AMBIENT ISOTOPY

Two knots K_1 and K_2 are called ambient isotopic if there exist a continuous family of homeomorphism $\{h_t : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \mid 0 \leq t \leq 1\}$ such that h_0 is an identity map. The function

$H : [0, 1] \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$H(t, p) = h_t(p)$ is continuous and $h_1(K_1) = K_2$.

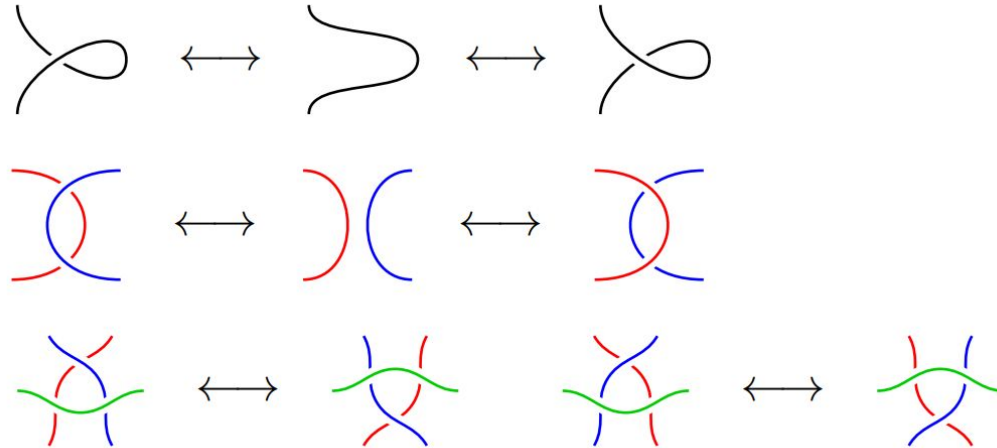
This captures the intuitive notion of two knots being same or of the same type.



Example of a complex looking unknot

REIDEMEISTER MOVE

A Reidemeister move is a local modification of a knot diagram that alters only a small neighbourhood of the diagram, leaving the rest unchanged. There are exactly three such moves.

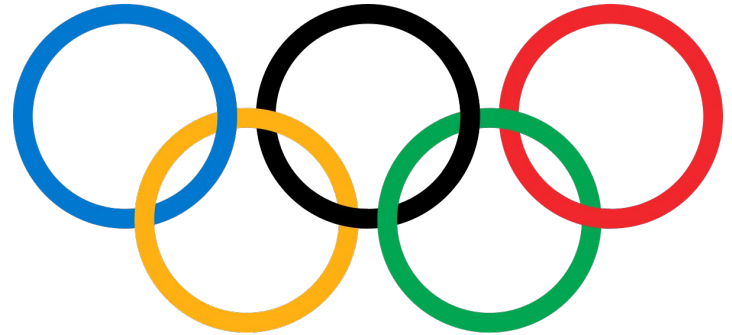
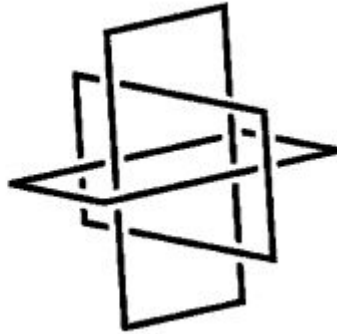
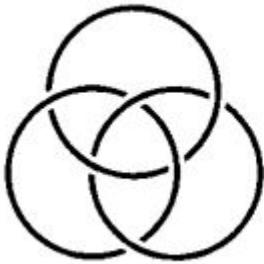


Three types of reidemeister moves

Reidemeister Theorem : Two diagrams of the same polygonal knot represent isotopic knots in $\mathbb{R}^3$ if and only if one diagram can be transformed into the other by a finite sequence of Reidemeister moves.

LINK

A Link is the disjoint union of a finite set of knots considered up to ambient isotopy. It is called *splittable* if it can be separated. A link is composed of component knots floating in 3D space. They cannot intersect, but they may be entangled with each other.

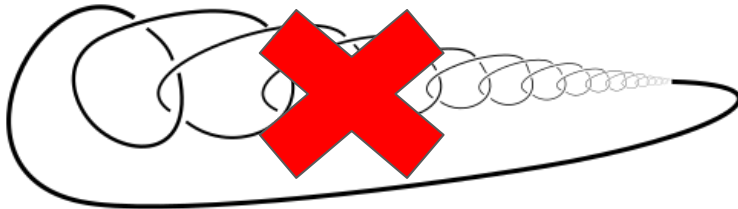


Two links are considered the same if one can be smoothly deformed into the other without cutting or passing strands through each other.

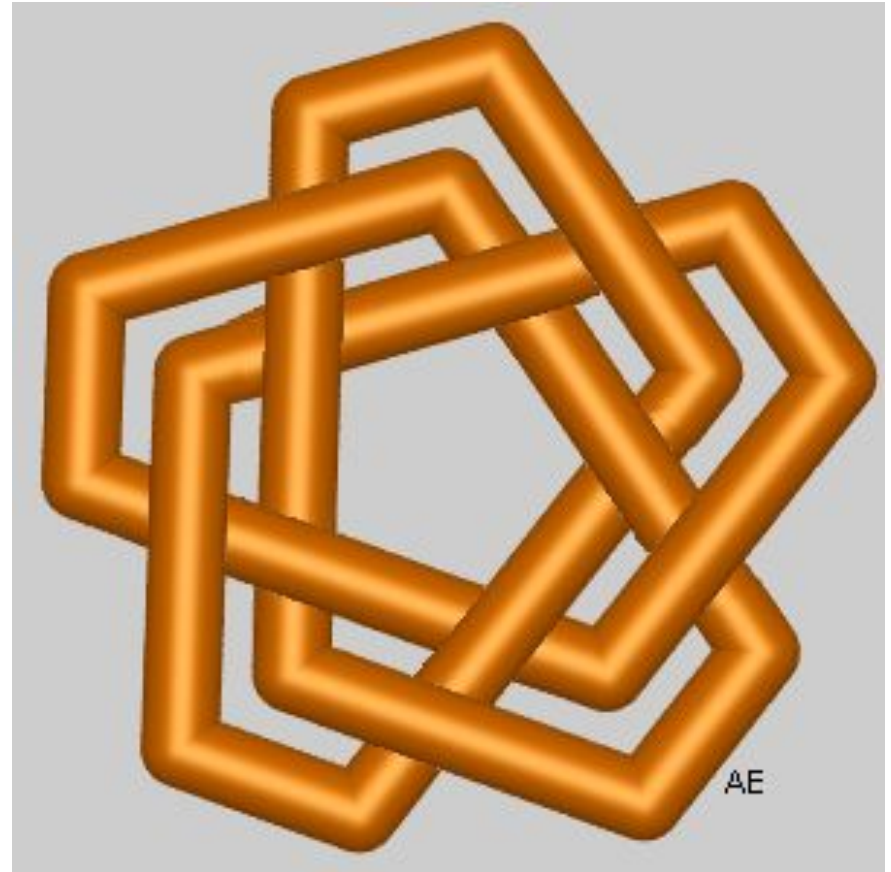
POLYGONAL & LATTICE KNOTS

Polygonal knots

A polygonal knot is the union of a finite number of closed straight-line segments called edges, whose endpoints are called the vertices of the knot. In knot theory we mostly discuss about tame knots, defined as knots which are equivalent to polygonal knots.

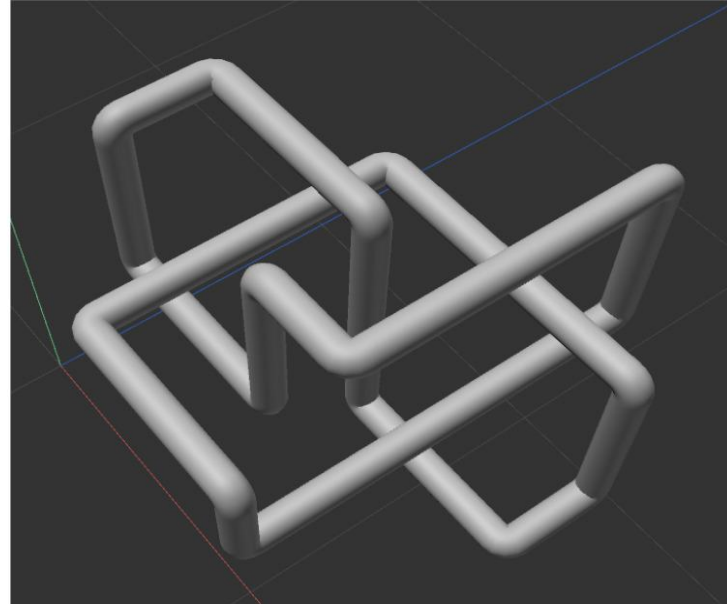
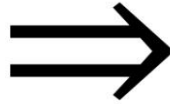
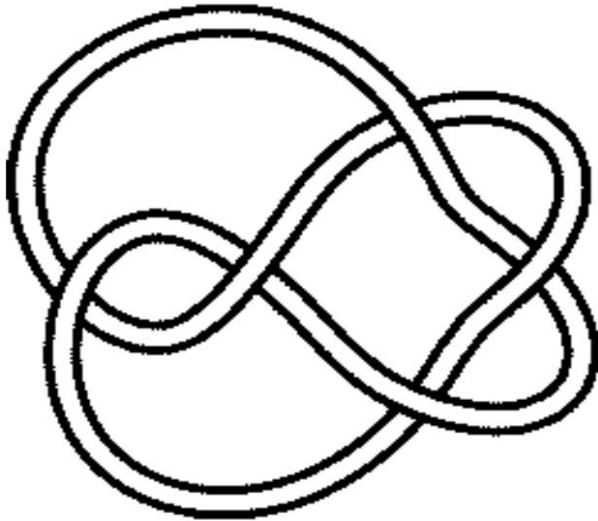


Example of Wild knot



LATTICE KNOTS

Lattice knots or cubulated knots in 3 – D are polygonal knots whose sides are parallel to one of the axes. It has been shown that every polygonal knot thus every tame knot is equivalent to a lattice knot .



KNOTS & WORDS

Words

Let us consider words over the alphabet $\{E, W, N, S, U, D\}$ with the following geometric meaning in mind:

$$E \leftrightarrow (1, 0, 0)$$

$$N \leftrightarrow (0, 1, 0)$$

$$U \leftrightarrow (0, 0, 1)$$

$$W \leftrightarrow (-1, 0, 0)$$

$$S \leftrightarrow (0, -1, 0)$$

$$D \leftrightarrow (0, 0, -1)$$

For example, NESW, NNESUWWDEESUW, and ϵ are words.

Displacement

For a word w , we define its displacement in $\mathbb{R}^3$ as:

$$D(w) = (|w|_E - |w|_W, |w|_N - |w|_S, |w|_U - |w|_D).$$

For a word $w = w_1 \dots w_{|w|}$, let us define the subword $w[i : j]$ for $1 \leq i \leq j \leq |w|$ as the contiguous segment $w_i \dots w_j$.

Further, let us call a subword $w' = w[i : j]$ *proper* iff $w' \neq w$ and $w' \neq \epsilon$.

Properties of Displacement

1. Additivity over concatenation

$$D(ww') = D(w) + D(w')$$

2. Subtractivity over substring deletion

$$D(ww_{sub}w') = D(ww') - D(w_{sub})$$

3. Permutation invariance:

For any $\pi \in S(|w|)$, define the permuted word $\pi(w)$ by $[\pi(w)]_i = w_{\pi(i)}$ for all $1 \leq i \leq |w|$

Then:

$$D(\pi(w)) = D(w)$$

Knotwords

Definition: We define a knotword w to be a word in the alphabet $\{N,S,E,W,U,D\}$ if it obeys the following geometric constraints:

1. $D(w) = \mathbf{0}$,

That is to say, the diagram specified by w is a closed loop.

2. no proper subword w' has $D(w') = \mathbf{0}$.

That is to say, the diagram specified by w does not contain any self-intersection.

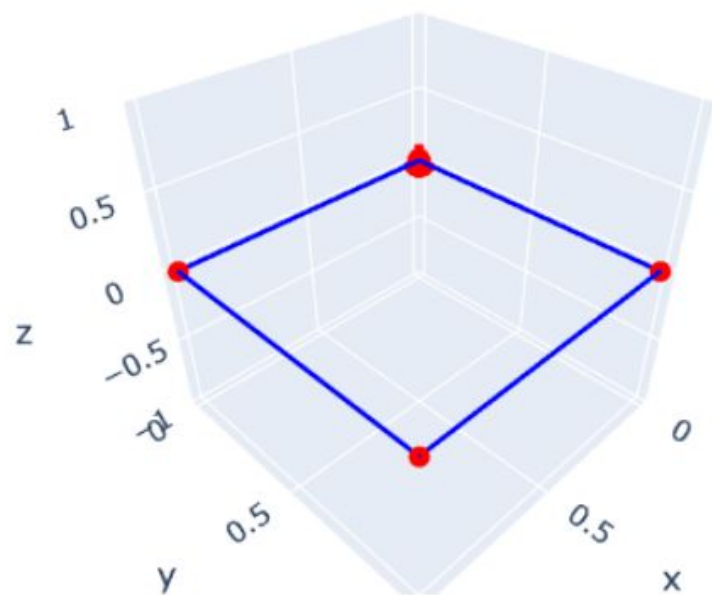


Figure 3.1: The knot NESW

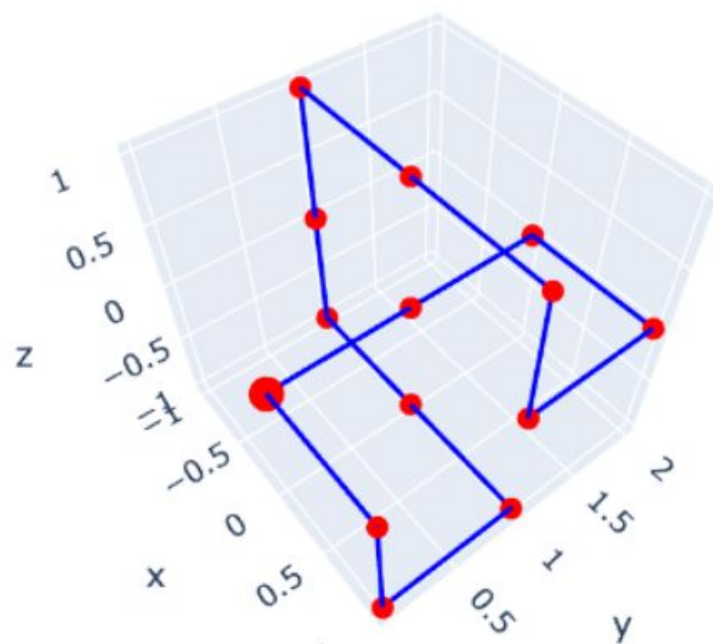
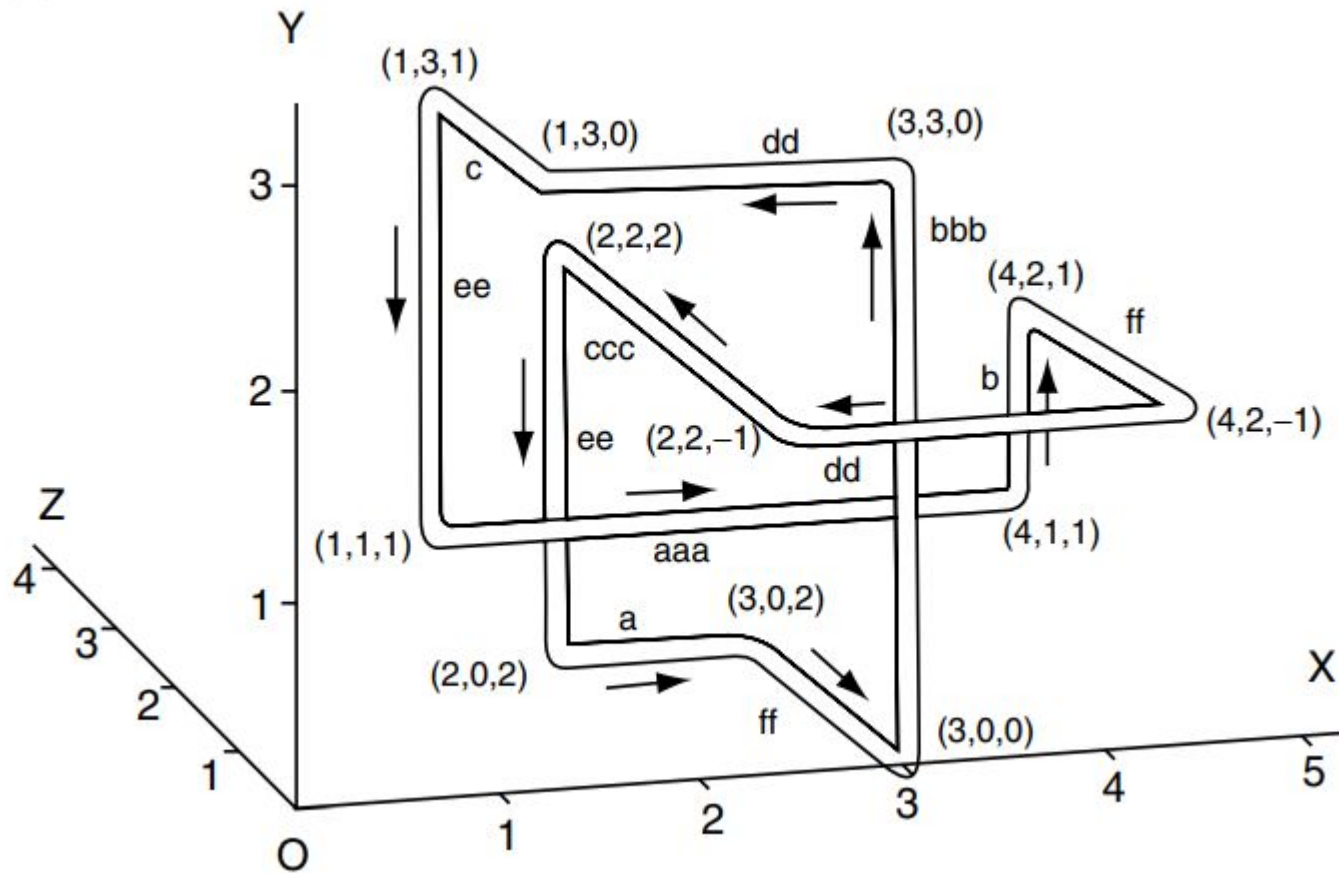


Figure 3.2: The knot NNESUWWDDEESUW

(b)



A Trefoil knot...

Elementary Moves

In 1954 Alan Turing enlisted all “elementary” moves that may be applied to a knotword and result in an isotopic knotword:

- (i) One may move a letter from one end of the row to the other.
- (ii) One may interchange two consecutive letters provided this still gives a knot.
- (iii) One may introduce a letter **E** in one place in the row, and **W** somewhere else, or **N** and **S**, or **U** and **D**, or take such pairs out, provided it still gives a knot.
- (iv) One may replace **E** everywhere by **EE** and **W** by **WW** or replace each **N** and **S** by **NN** and **SS** or each **U** and **D** by **UU** and **DD**. One may also reverse any such operation.

Doubling

Definition: Doubling along X (or analogously Y, or Z)

Doubling along the X-axis is the application of string homomorphisms

$$E \mapsto EE \text{ along with } W \mapsto WW$$

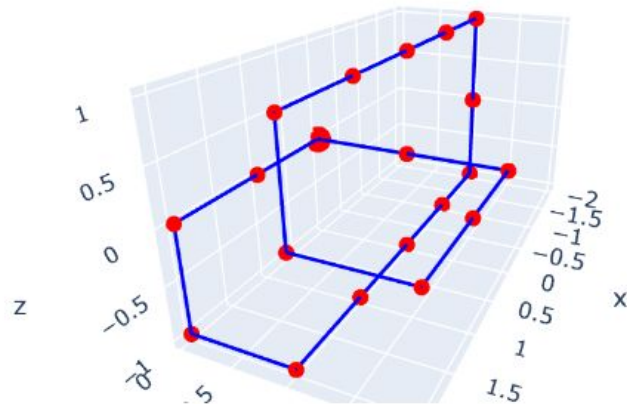
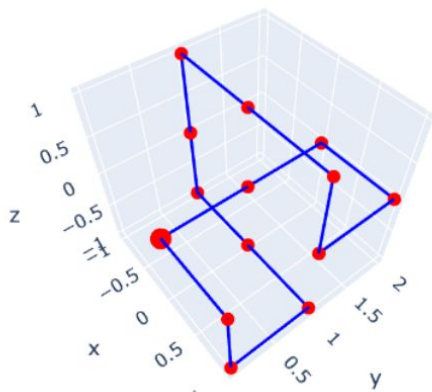
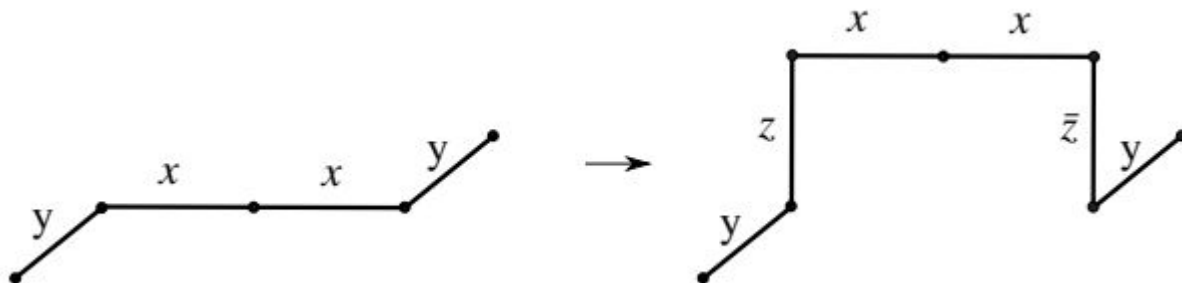


Figure 3.2: The knot N NESUWWDDEESUW

Elementary Switch

Definition: Elementary Switch

An elementary switch of a word w is the replacement of a subword w' by $Aw'B$ where $D(AB) = 0$ followed by recursive reduction according to $NS \mapsto \epsilon$, $EW \mapsto \epsilon$, $UD \mapsto \epsilon$.



For example:

$NEUWS \rightarrow NE\underline{U}DWS\underline{U} \rightarrow \underline{NE}WS\underline{U} \rightarrow \underline{NS}\underline{U} \rightarrow \underline{U}$

Elementary Move Theorem

Theorem (Anan'in, Grishkov, Korshunov): Every *Elementary Move*, as put forward by Turing, is achievable using Elementary Switches alone.

Thus, Knotword Isotopy is solely a matter of

- applying elementary switches and
- reducing.

Elementary Moves

In 1954 Alan Turing enlisted all “elementary” moves that may be applied to a knotword and result in an isotopic knotword:

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The Question of Knot Theory

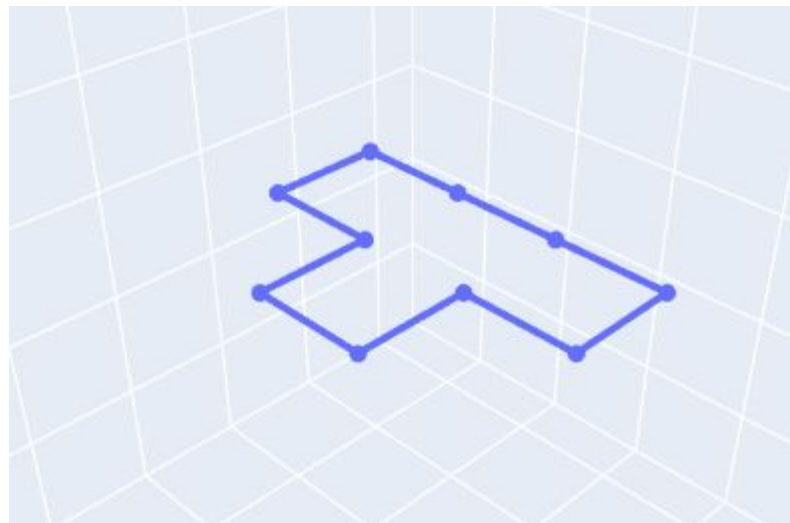
The main problem Knot Theory serves to answer is the question of whether two knots are equivalent under ambient isotopy.

A historically relevant subproblem is the [Unknotting] Problem: Decide whether a given knot is equivalent to the unknot.

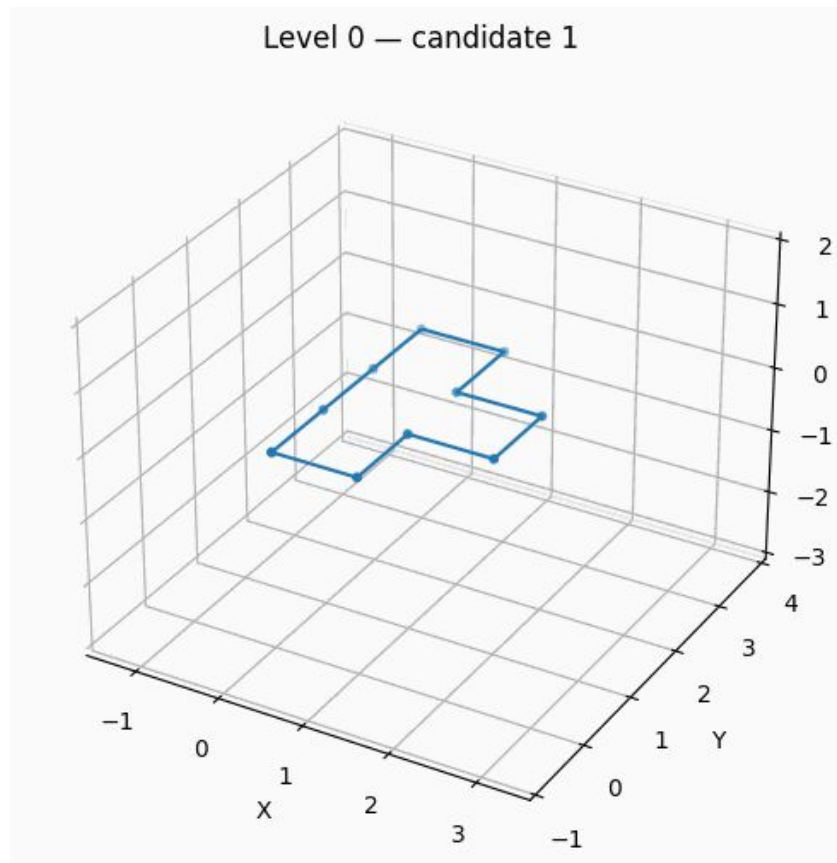
[Unknotting] is in NP. It is unknown whether it is NP Complete.

We coded an “Acceptor” machine in Python that leverages the Elementary Move Theorem to breadth-first explore the search space of a knotword.

If it accepts the target knotword as equivalent, it terminates, in exponential time.



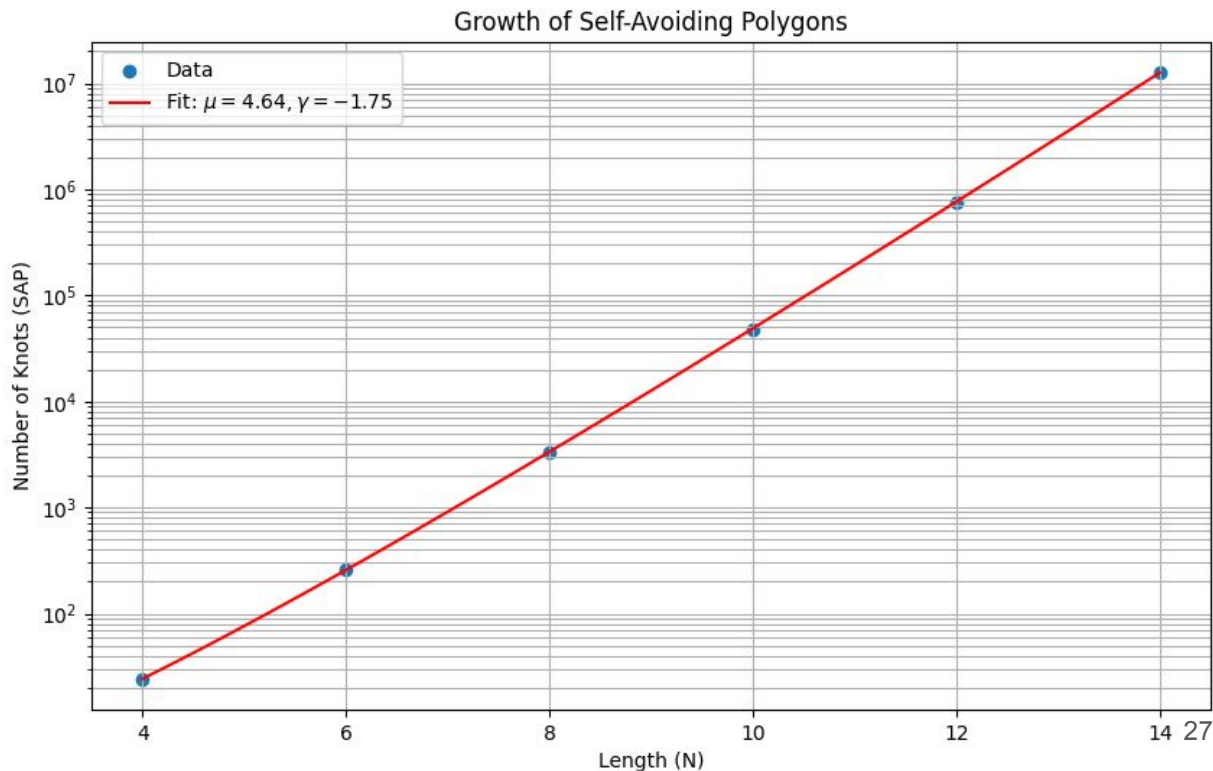
Starting lattice knot



Knot Counts

We brute forced the generation of knots of length up to 14. The counts were found to grow exponentially according to $C(n) = A\mu^n n^\gamma$

	knot_counts.csv
1	Length,Count
2	4,24
3	6,264
4	8,3312
5	10,48240
6	12,762096
7	14,12673920
8	



Lower-Bound For Length of Non-trivial Lattice Knot

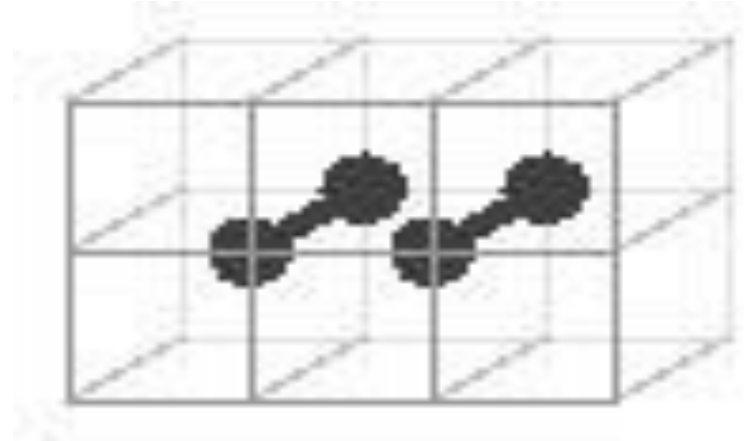
We know by enumerating all the knots of length lesser than 24 that 24 is the minimum length needed for making a non-trivial knot (trefoil).

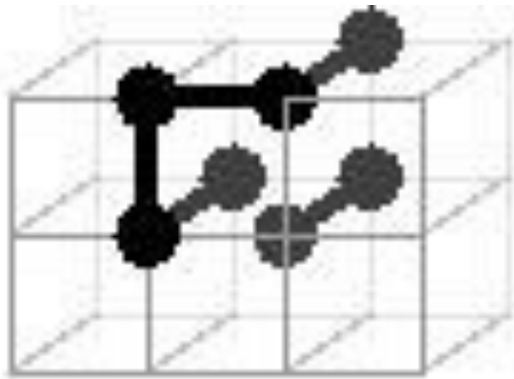
Here we would prove that it must be at least 20 .

- Minimal Bounding Box
- Assume a box B bounding any non-trivial knot and let this knot be one with minimum length .
- For Minimal bounding box of size $(1, 1, n)$, intersection of the knot and the interior of the box is empty thus the knot lie on the surface of the Box .
- Any two simple polygonal arcs in S^2 with the same endpoints are ambient isotopic relative to the endpoints.

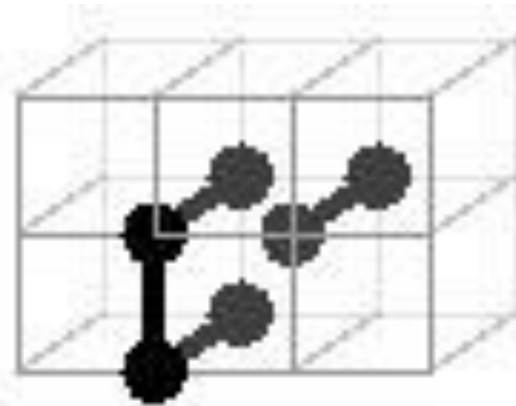
Continued..

- For $(2, 2, 2)$ either the intersection is empty or it is one of the two following cases
 - a line through the center of two opposite faces
 - union of two lines from center of two adjacent faces to the center of the cube.
- For $(1, 2, 3)$ if the knot is non trivial then the interior must have two edges .
- Consider the intersection J of the knot with the near side of the box, i.e., the face nearer to the camera in the picture.
- J must cannot have length 1 as it will create a U-turn like structure which breaks the assumption of it being of minimum length.

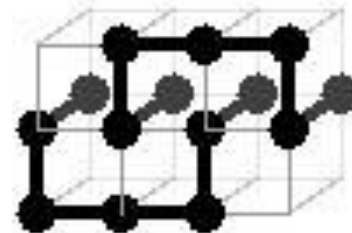
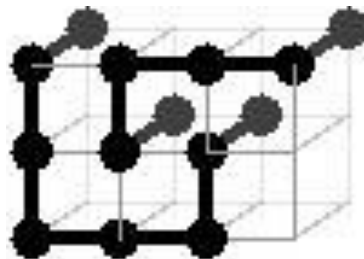
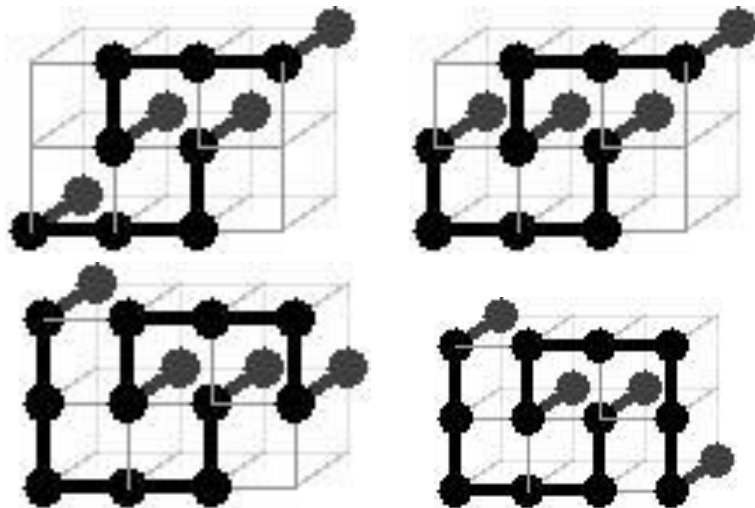




For length of $J = 2$



If length of J is 1



Cases for $J \geq 3$

Continued.....

- All of these cases can be verified to wind up as an unknot or to get us a link (the Hopf link).
- So we showed that if the minimum bounding box B of a non-trivial knot has size (a, b, c) then $(a+b+c) \geq 7$.
- **Now we prove that a knot bounded by the minimal bounding box $B(a,b,c)$ has length greater than or equal to $2(a + b + c) + 6$.**
- The range of K along the x -axis is a . We claim the knot has length at least $2a$ along the x -axis, by virtue of being a closed loop.

Proposition 4.4.1. *Let K be a cubulated knot with knotword w . If the range of K along the X -axis is a , then $|w|_E + |w|_W \geq 2a$. Analogous statements follow for Y , and Z .*

- Similarly for y and z axis we have length $2b$ and $2c$ at least.

Continued...

Let's now show that if this value is exactly $2a$ that means if one starts at a point with minimum x -coordinate then the displacement function along the x -axis will be a function which is non-decreasing at first until it reaches the maximum value a , and then it decreases to 0.

We prove by induction on a that any such lattice knot is an unknot.

Base Case : For $a = 1$, the knot has exactly two edges parallel to the x -axis say AB , CD . Say A and B lie on $x = 0$. In each of these planes, the intersection of the knot with the plane is a simple arc connecting the two endpoints A and B . Each piecewise linear planar arc can be to the straight segment joining its endpoints. Therefore, the entire knot can be a deformed to a rectangle.

Inductive step

- Examine the plane $x = a - 1$.
- There are exactly two points P and Q where the edges parallel to the x -axis connecting $x = a - 2$ to $x = a - 1$ meets this plane .
- Now consider the intersection of the knot with the plane $x = a$. This intersection is a simple arc connecting two endpoints lying on that plane. Since any two simple arcs in a plane with fixed endpoints are ambient isotopic, we may choose within the plane $x = a$ a simple polygonal arc connecting these endpoints whose projection to the yz -plane does not intersect the projection of the portion of the knot lying in the plane $x = a - 1$. This is possible because a finite collection of line segments in a plane cannot disconnect the plane unless they form a closed curve; in our situation the knot is not entirely contained in the yz -plane, and it cannot have any self-intersections, so such a separating closed curve cannot occur.

Continued...

- Having chosen such an arc in $x = a$, we now shorten the two x -parallel edges that join the planes $x = a - 1$ and $x = a$, continuously sliding the planar arc from $x = a$ toward $x = a - 1$. Since this deformation does not introduce intersections, we obtain an ambient isotopy that pushes the entire portion of the knot lying in $x = a$ into the plane $x = a - 1$. Thus, after the deformation, the knot lies entirely in the slab $0 \leq x \leq a - 1$ and uses only $2(a-1)$ edges parallel to the x -axis. The resulting knot is therefore, by the inductive hypothesis, isotopic to the unknot. So length along x -axis is at least $2a + 2$ for non trivial knot. Applying the same result for all the axes, we can prove our claim. Put $a + b + c \geq 7$ and we will get our lower bound of 20.

FREE GROUPS

WORDS

Let A be a given alphabet. A syllable is formal expression a^n ; $a \in A, n \in \mathbb{Z}$

A word is a finite sequence of syllables written as

$$a_1^{n_1} a_2^{n_2} \dots a_k^{n_k} = \prod_{i=1}^k a_i^{n_i}$$

Empty word or the word with no syllables is denoted by 1 .

In the set $W(A)=W$ define a binary operation called multiplication where if

$$u = a_1^{n_1} a_2^{n_2} \dots a_k^{n_k} \text{ and } v = b_1^{m_1} b_2^{m_2} \dots b_l^{m_l}$$

$$\text{Then } uv = a_1^{n_1} a_2^{n_2} \dots a_k^{n_k} b_1^{m_1} b_2^{m_2} \dots b_l^{m_l}$$

We can verify that W with this operation is a semigroup

Equivalence and Contraction

On $W(A)$ we define four elementary operations

Type 1 contraction or expansion : $wa^0w' \leftrightarrow ww'$

Type 2 contraction or expansion : $wa^p a^q w' \leftrightarrow wa^{p+q} w'$

We define an equivalence relation on the words such that two words are equivalent if one can be obtained from others by a finite sequence of elementary operations. The set $F[A]$ be the set of equivalence classes of words in $W[A]$. We can prove

if $u \sim u'$ and $v \sim v'$. Then $uv \sim u'v' \sim u'v$.

So in $F[A]$ we can define a binary operation multiplication defined as

$\alpha, \beta \in F[A]$ and $\alpha = [u]$ and $\beta = [v]$ then $\alpha\beta = [u][v] = [uv]$.

u^{-1} such that if $u = a_1^{n_1} a_2^{n_2} \dots a_k^{n_k}$ then $u^{-1} = a_k^{-n_k} a_{k-1}^{-n_{k-1}} \dots a_1^{-n_1}$

Notice uu^{-1} and $u^{-1}u$ are both equivalent to 1.

$F[A]$ is a group under the aforementioned operation

WORD PROBLEM FOR FREE GROUP

To work with $F[A]$ first we have to decide if two words are equivalent or not.

We need an algorithm for this task .

We first define what a reduced word is . A reduced word is a word where one cannot apply any elementary operations .

Define a function on $W(A)$ where $\rho(w) = w$ if w is reduced otherwise, start iterating w and apply elementary contraction on the first syllable where it is possible to apply one.

Consider the sequence $w, \rho(w), \rho^2(w), \dots$. Choose the first r such that

$$\rho^{r+1}(w) = \rho^r(w) \text{ Denote } w^! = \rho^r(w) .$$

Property of reduced words

1. $\rho(w) = w$ if and only if w is reduced.
2. If u is not reduced then $\rho(uv) = \rho(u)v$.
3. $w \sim w^!$ and $w^!$ is reduced.
4. w is reduced if and only if $w = w^!$
5. $u \sim v$ if and only if $u^! = v^!$.

Presentation of Groups

To naively store a finite group in computer memory, one needs to save:

1. **N elements** for $G = \{g_1, g_2, \dots, g_n\}$
2. **N^2 elements** for the operation $*: G \times G \rightarrow G = (a, b) \mapsto a*b$

Which is essentially a Cayley table. This a space-inefficient depiction of groups.

	1	a	b
1	1	a	b
a	a	b	1
b	b	1	a

What is a presentation

Recall free groups on a generator set S are

- $S=a$: $\langle a \rangle$ is the free group on $\{a\}$, the set of all integer powers of a
- $S=a,b$: $\langle a,b \rangle$ is the free group that consists of all finite products of a,b
- Etc

For a subset R of $\langle a,b \rangle$, we use the notation $\langle a,b \mid R \rangle$ to represent the homomorphic image of $\langle a,b \rangle$ under a map with kernel generated by R . In other words this is the factor group of $\langle a,b \rangle$ modulo the normal closure of R .

These 'relations' R capture equations of element interactions in the group

For example $\langle a,b \mid aba^{-1}b^{-1} \rangle$ is the free group on 2 generators with $aba^{-1}b^{-1}=e$

Examples

1. $\langle a, b \mid aba^{-1}b^{-1} \rangle$ or for clarity $\langle a, b \mid ab=ba \rangle$ is an instruction to construct all finite products of a, b , and then reduce them according to the rule $ab=ba$. This is just $\mathbb{Z}^2$
2. $\langle a \mid a^3 \rangle$ takes the cyclic group $\langle a \rangle$ and reduces it according to

$$a^m = a^{m \pm 3} = a^{m \pm 6} = \dots = a^{m \bmod 3}$$

This is just $\mathbb{Z}_3$

3. $\langle a, b \mid f(a, b) \rangle = \langle a, ab \mid f(a, b) \rangle = \langle ba, b \mid f(a, b) \rangle = \langle a, b, c \mid f(a, b), c=g(a, b) \rangle$
4. $\langle a, b \mid f(a, b), g(a, b) \rangle = \langle a, b \mid f(a, b), g(a, b), f(a, b)^n g(a, b)^m \rangle$

Tietze Theorem

The Tietze theorem of equivalent presentations formalizes our intuition about the generator-relation presentation of groups. Any presentations of the same group are always linked by finitely many steps of the four fundamental moves:

Type 1: For a redundant relator r that lies in the normal closure of the remaining, it may be added/removed from the presentation. For such an r ,

$$\langle \mathbf{X} \mid \mathbf{R}, r \rangle = \langle \mathbf{X} \mid \mathbf{R} \rangle$$

Type 2: A redundant generator may be added/removed. Precisely,

$$\langle \mathbf{X} \cup \{y\} \mid \mathbf{R}, y=f(\mathbf{X}) \rangle = \langle \mathbf{X} \mid \mathbf{R} \rangle$$

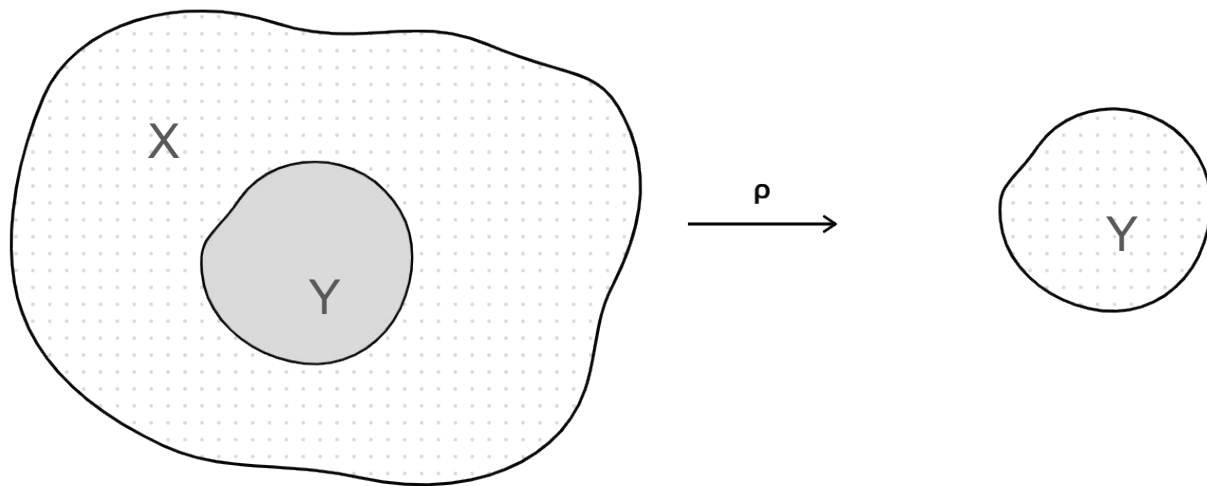
Example of Tietze Theorem: $\langle x, y, z \mid xyz = yzx \rangle = \langle x, y, a \mid xa = ax \rangle$

$$\begin{array}{l}
 (x, y, z : xyz(yzx)^{-1}) \\
 \downarrow \mathbf{II} \\
 (x, y, z, a : xyz(yzx)^{-1}, a(yz)^{-1}) \\
 \downarrow \mathbf{I} \\
 (x, y, z, a : xa(ax)^{-1}, a(yz)^{-1}, xyz(yzx)^{-1}) \\
 \downarrow \mathbf{I}' \\
 (x, y, z, a : xa(ax)^{-1}, a(yz)^{-1}) \\
 \downarrow \mathbf{I} \\
 (x, y, z, a : xa(ax)^{-1}, z(y^{-1}a)^{-1}, a(yz)^{-1}) \\
 \downarrow \mathbf{I}' \\
 (x, y, a, z : xa(ax)^{-1}, z(y^{-1}a)^{-1}) \\
 \downarrow \mathbf{II}' \\
 (x, y, a : xa(ax)^{-1}).
 \end{array}$$

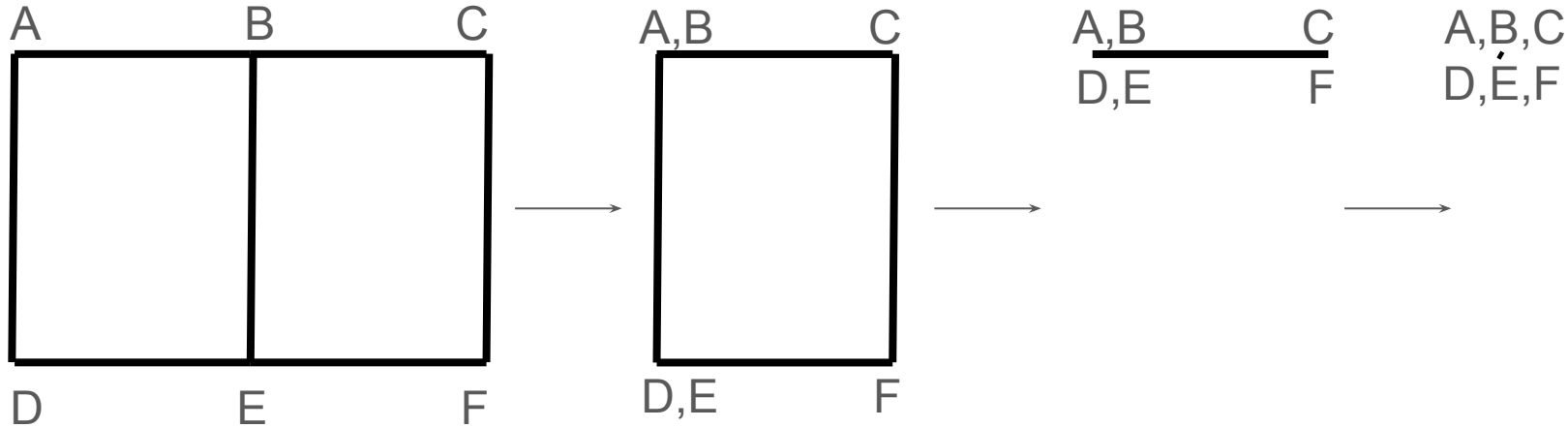
FUNDAMENTAL GROUP

Retraction

Definition: A retraction of a topological space X onto a subspace $Y \subseteq X$ is a continuous mapping $p : X \rightarrow Y$ such that for any $p \in Y$, $p(p) = p$. A subspace Y is called a retract of X if there exists a retraction of X onto Y .



Retractions can cause 'loss of holes'



We don't imagine retractions in X to play like a movie from $t=0$ to $t=1$.

Retraction: Theorem

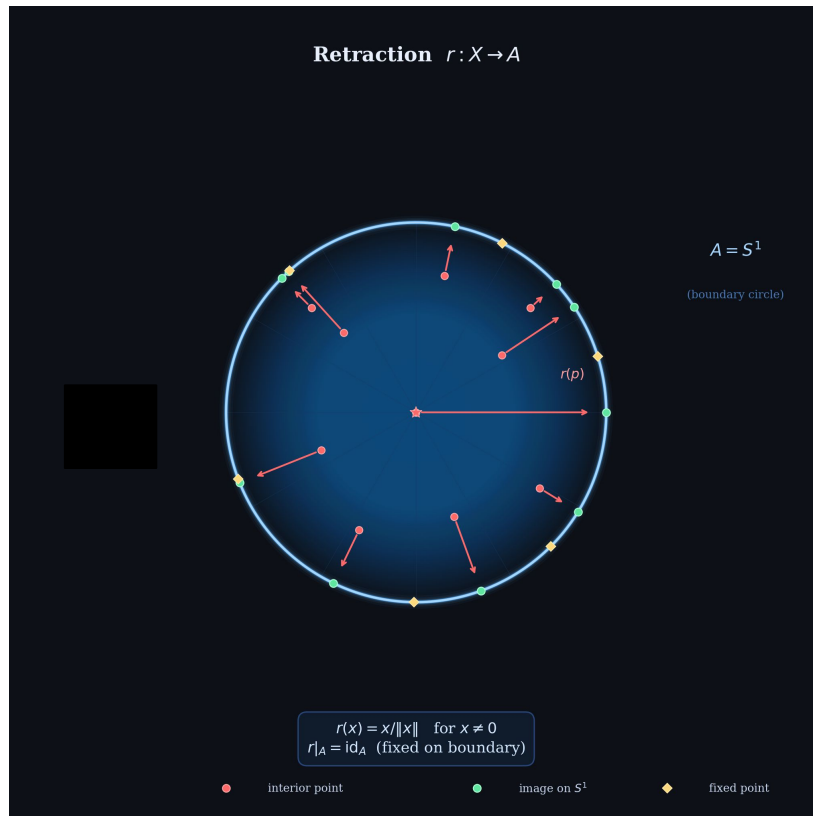
Theorem : If $\rho : X \rightarrow Y$ is a retraction and X is path-connected, then for any basepoint $p \in Y$, the induced homomorphism $\rho_* : \pi_1(X, p) \rightarrow \pi_1(Y, p)$ is surjective.

This formalizes the intuition that holes may be lost but never created.

Application:

This theorem proves why a solid disc cannot retract $\pi_1(D^2) \cong \{1\}$ onto $\pi_1(S^1) \cong \mathbb{Z}$, retraction existed, it would induce a surjection from

Which is impossible.



Retraction to a point is a trivial retract that exists for every simply connected space and any point in it

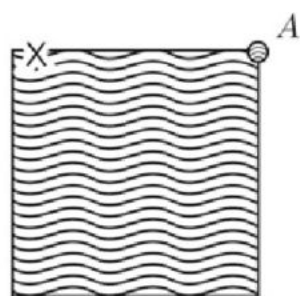
$$X = D^2 \setminus (0,0) \rightarrow Y = S^1$$

Deformation

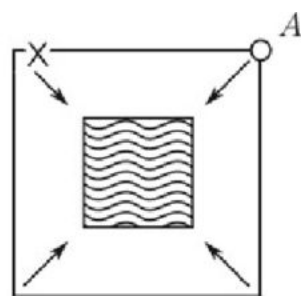
Definition : A deformation of a topological space X is a family of continuous mappings $h_s : X \rightarrow X$, for $s \in [0, 1]$, such that h_0 is the identity map on X , and the map $H : X \times [0, 1] \rightarrow X$ defined by $H(x, s) = h_s(x)$ is continuous.

A space X is *deformable into* a subspace Y if there exists a deformation $\{h_s\}$ such that the image $h_1(X)$ is contained in Y . If $h_1(X) = Y$, we say X is *deformable onto* Y .

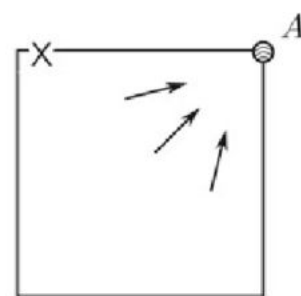
Theorem : If $\{h_s\}$, $0 \leq s \leq 1$, is a deformation of X , then for any basepoint $p \in X$, the induced homomorphism $(h_1)_* : \pi_1(X, p) \rightarrow \pi_1(X, h_1(p))$ is an isomorphism.



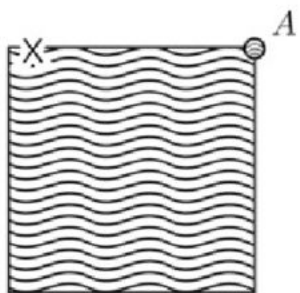
$t = 0$



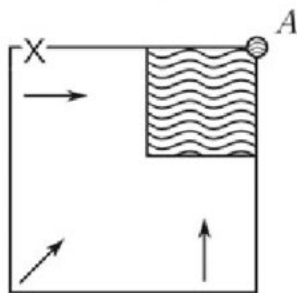
$t = \frac{1}{2}$



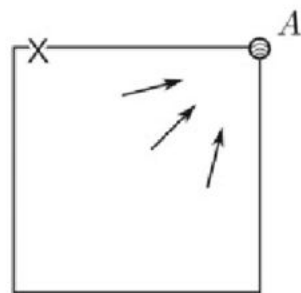
$t = 1$



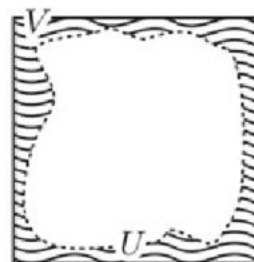
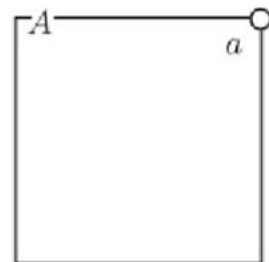
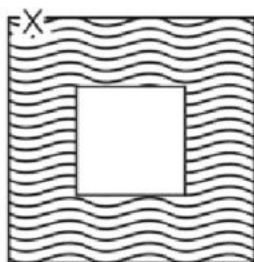
$t = 0$



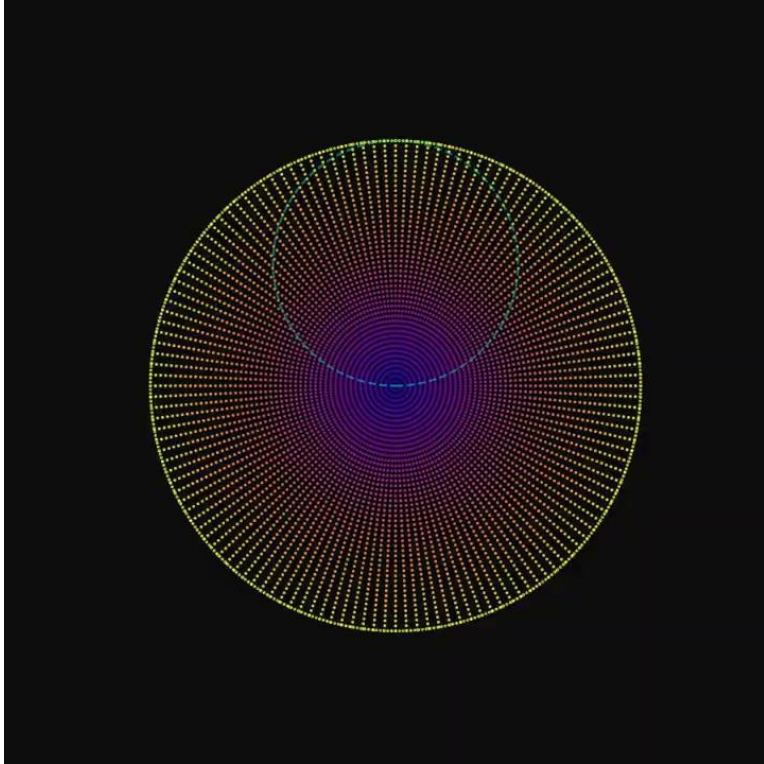
$t = \frac{1}{2}$



$t = 1$



Example of Deformation: Disc to boundary



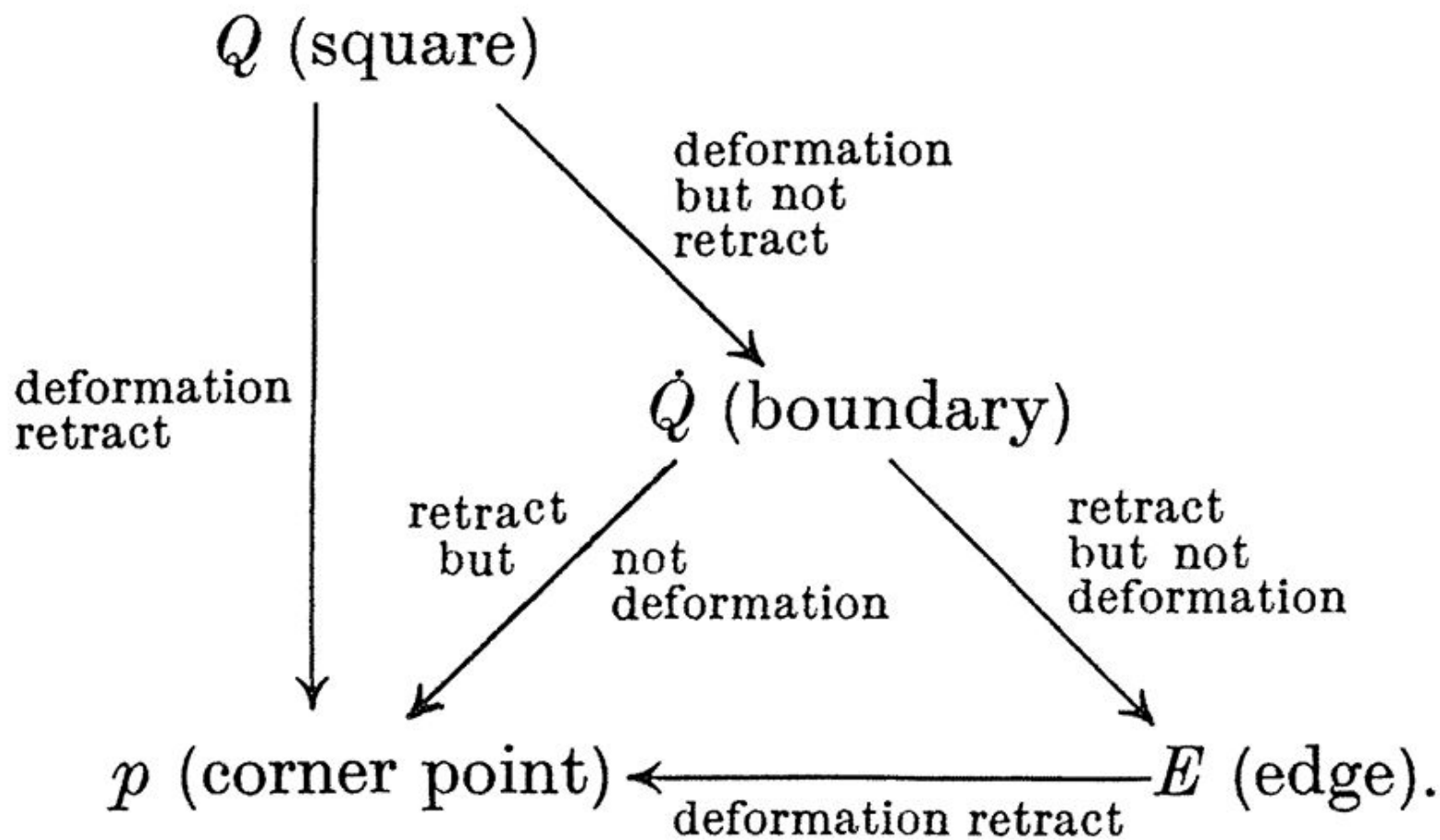
- As opposed to retractions, we imagine deformations to play like a movie from $t=0$ to $t=1$.
- Deformation of a space may create holes

Retraction and Deformation

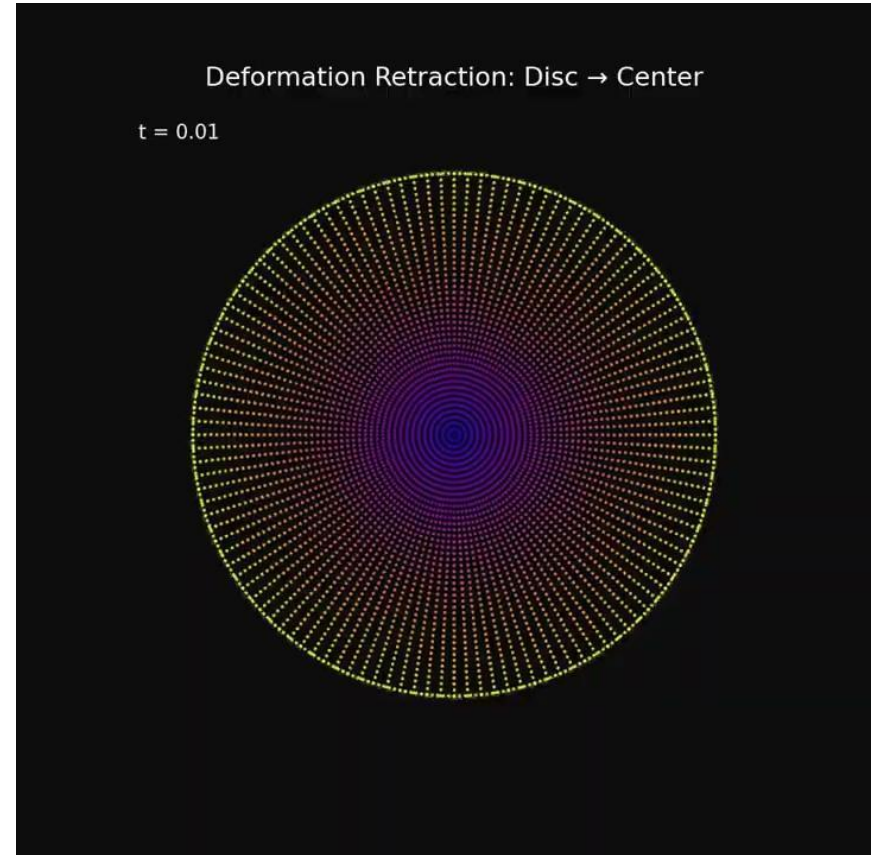
Theorem: If a continuous mapping $f : X \rightarrow Y$ is realizable by a deformation of X , then the induced homomorphism $f_* : \pi_1(X, p) \rightarrow \pi_1(Y, f(p))$ is an injective homomorphism.

Definition: A subspace $Y \subseteq X$ is a deformation retract of X if there exists a retraction $p : X \rightarrow Y$ which is realizable by a deformation of X . Equivalently, there exists a deformation $\{h_s\}$ of X such that $h_1(X) \subseteq Y$ and $h_s(y) = y$ for all $y \in Y$ and $s \in [0, 1]$.

Theorem: If a subspace Y is a deformation retract of a path-connected topological space X , then the inclusion map $i : Y \rightarrow X$ and the retraction map $p : X \rightarrow Y$ induce mutually inverse isomorphisms between $\pi_1(X, p) \cong \pi_1(Y, p)$ $p \in Y$. Consequently $\pi_1(X) \cong \pi_1(Y)$.



For simply connected spaces, the trivial retract to a point can always be made into a deformation retract

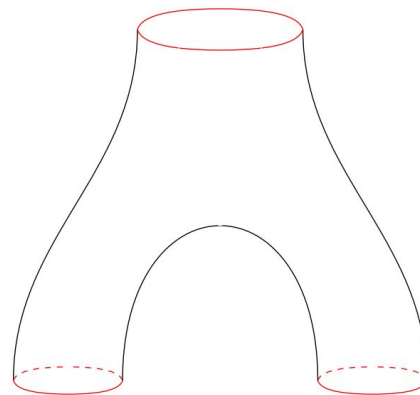
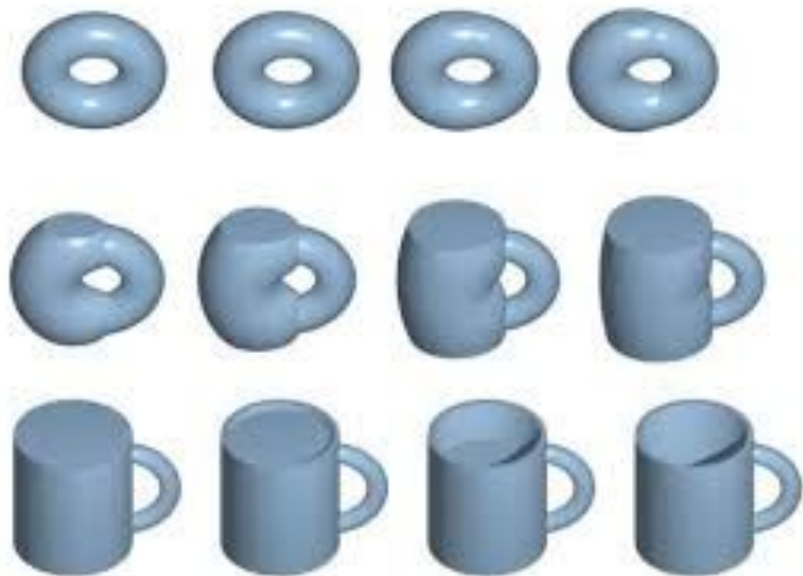


Homotopy Type

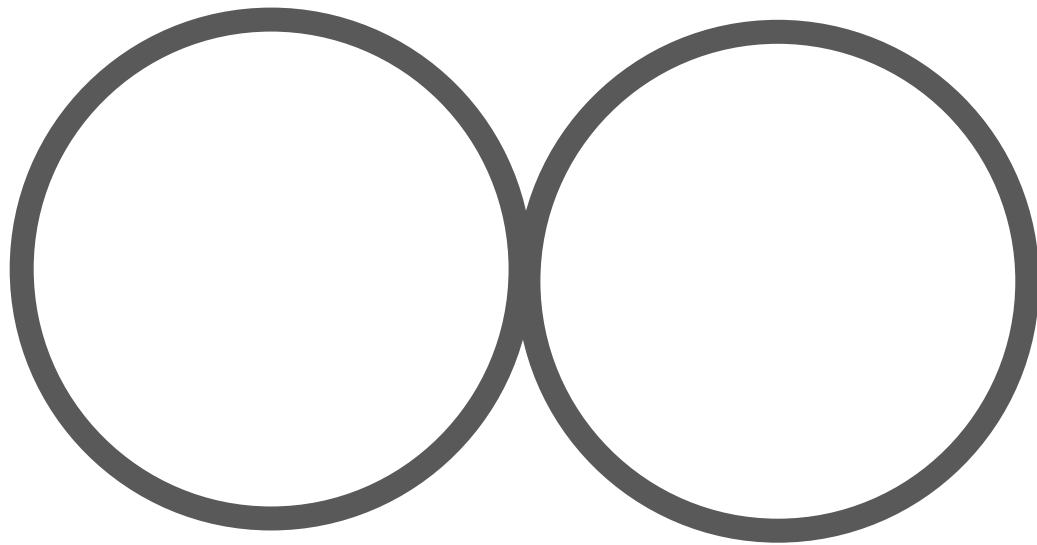
Homotopy Type : Two topological spaces X and Y are of the *same homotopy type* (or are *homotopy equivalent*) if there exists a finite sequence of topological spaces $X = X_0, X_1, \dots, X_n = Y$ such that for each $i = 1, \dots, n$, either X_i is homeomorphic to X_{i-1} , or X_i is a deformation retract of X_{i-1} , or vice versa.

Theorem: If X and Y are path-connected spaces of the same homotopy type, then $\pi_1(X)$ is isomorphic to $\pi_1(Y)$.

Examples



Fundamental Group of 2-Leafed Rose



The fundamental group is $F[a,b]$

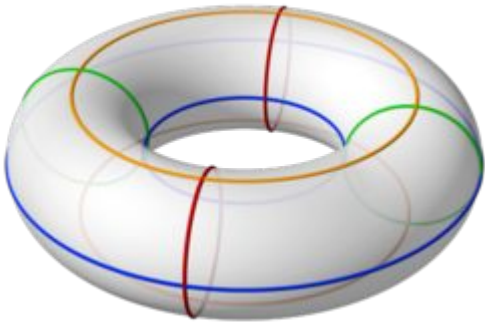
Any finite product
 $a^l b^m a^n b^p \dots$

Corresponds to a
path and its
homotopy class

Fundamental Group of Torus

One might naively repeat the same argument as last time, namely assigning “a” to latitude paths and “b” to longitude paths.

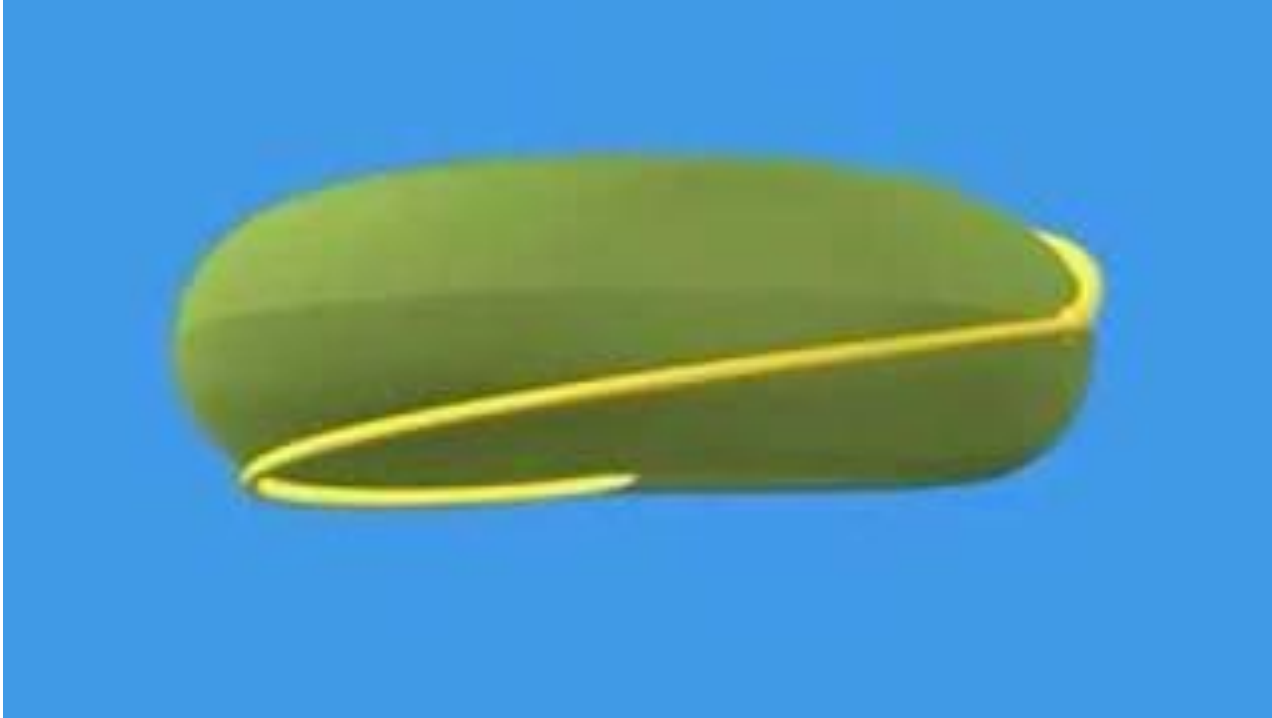
Then the group of path homotopy classes i.e. the fundamental group seems to be $F[a,b]$ again.



However, the path ab and the path ba are homotopic to each other.

It requires a little visualization

Latitude and Longitude of Torus are commutative



Fundamental Group of Torus II

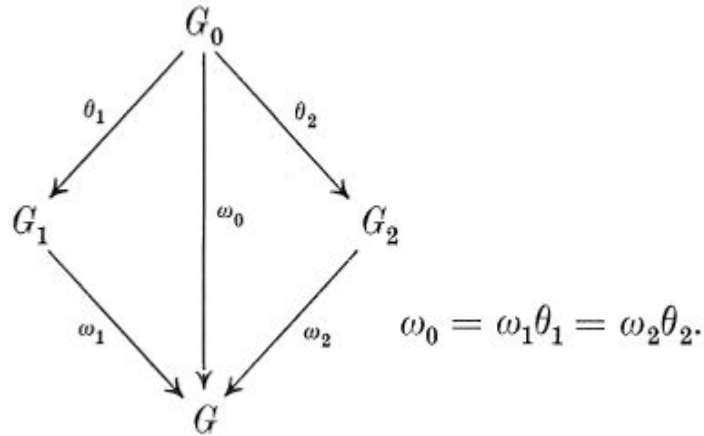
Thus the fundamental group of a torus is

$$\langle a, b \mid ab = ba \rangle \cong \mathbb{Z}^2$$

Is there a proof that $ab=ba$ is the only relation on a,b ?

Is there a general procedure to compute the fundamental group of a surface based on already identified loop-classes on it? That is, having identified the possibly dependent generators, a theorem that declares all the relations between them

Van Kampen Theorem



$$G_1 = \pi(X_1)$$

$$G_2 = \pi(X_2)$$

$$G_0 = \pi(X_1 \cap X_2)$$

$$G = \pi(X_1 \cup X_2)$$

Given $X_1, X_2, X_1 \cap X_2$ are open and path connected

Seifert-van Kampen Theorem:

The image groups $\omega_1(G_1)$ and $\omega_2(G_2)$

generate G . Furthermore, if H is an arbitrary group and $\psi_i : G_i \rightarrow H$ (for $i = 1, 2$) are homomorphisms satisfying the consistency condition $\psi_1 \circ \theta_1 = \psi_2 \circ \theta_2$, then there exists a unique homomorphism $\lambda : G \rightarrow H$ such that $\psi_1 = \lambda \circ \omega_1$ and $\psi_2 = \lambda \circ \omega_2$.

Corollaries of van Kampen theorem

If X_1 and X_2 are simply-connected, then so is $X = X_1 \cup X_2$.

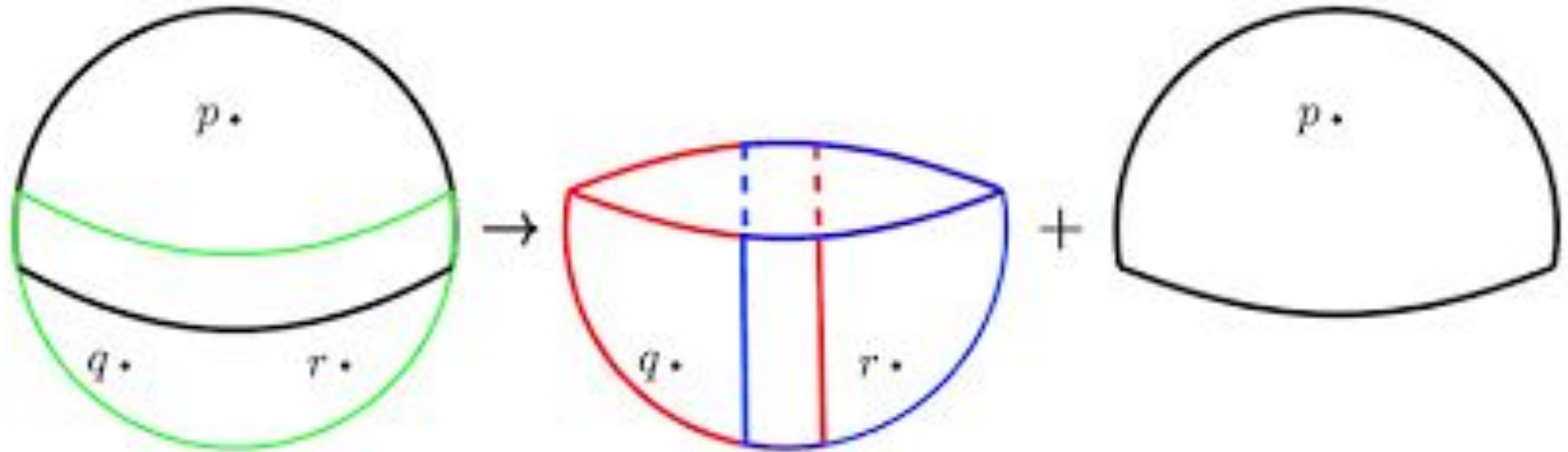


Fig : Van kampen for sphere

N-leafed rose

Corollary : *If G_0 is trivial, and G_1, G_2 are free groups with free bases $\{\alpha_1, \alpha_2, \dots\}$ and $\{\beta_1, \beta_2, \dots\}$ respectively, then G is free, and the union $\{\omega_1(\alpha_1), \dots, \omega_2(\beta_1), \dots\}$ is a free basis for G .*

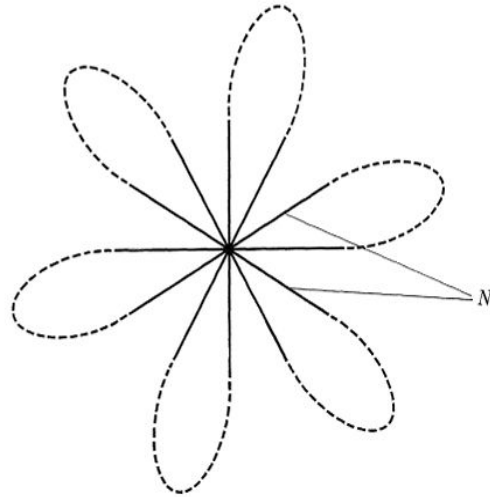
$C_{(n)}$ – union of n topological circles joined at p

N – an open neighbourhood of p in $X = C_{(n+1)}$

$$X_1 = C_{(n)} \cup N$$

$$X_2 = X_{n+1} \cup N$$

$$X_1 \cap X_2 = N$$



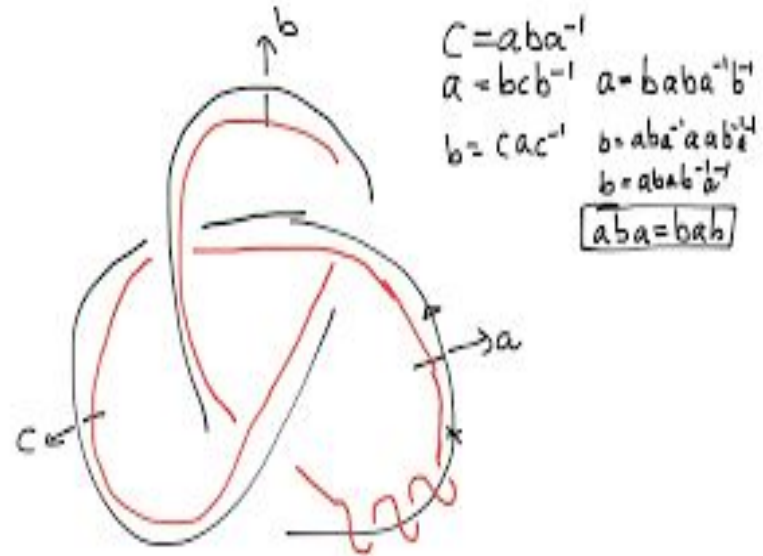
PRESENTATION OF KNOT GROUP

The Knot Group

Let $K \subset \mathbb{R}^3$ be a knot and p_0 be a point on $\mathbb{R}^3 \setminus K$. The knot group of K is defined as $G(K) = \pi(\mathbb{R}^3 \setminus K, p_0)$.

As $\mathbb{R}^3 \setminus K$ is path connected, for all values of p_0 the groups will be isomorphic.

We are going to study two different presentations of a knot which are isomorphic to each other, namely over presentation and Wirtinger presentation.



Over Presentation

Choose a set of $2n$ points Q which divide the knot in $2n$ connected segmented arcs each of which is either an underpass and overpass such that they are alternated around the knot.

We can orient the knot so that the points Q lie on the XY plane , the overpasses over it and the underpasses under it .

Choose a point P_0 over the knot on the z axis and a q_0 on $R^2 \setminus P(K)$

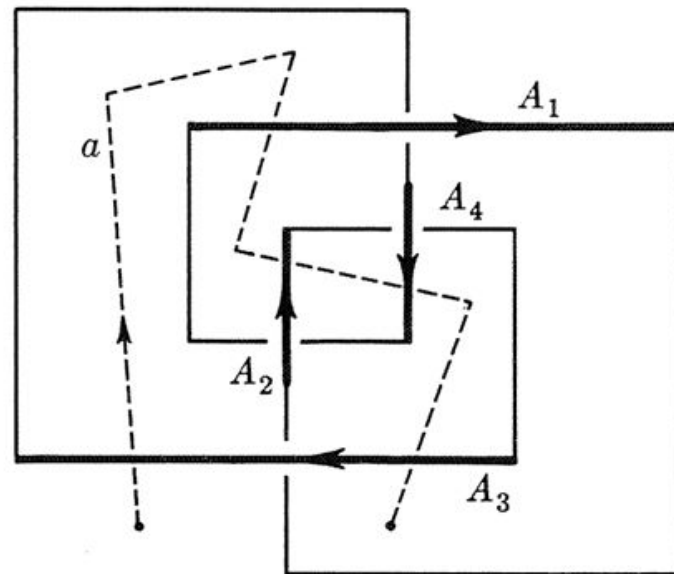


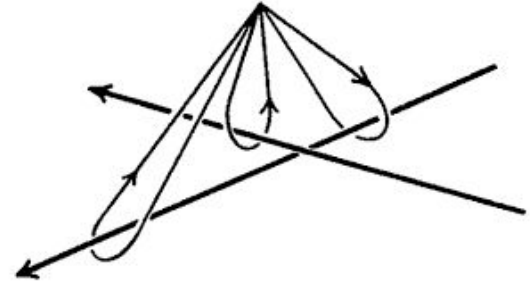
Figure 28. $a^\# = x_3 x_1 x_2 x_4^{-1} x_3^{-1}$

Over Presentation

Let A_i be the underpasses and B_i be the overpasses for $1 \leq i \leq n$

$$A = A_1 \cup A_2 \cup \dots \cup A_n$$

$$B = B_1 \cup B_2 \cup \dots \cup B_n$$



Choose any polygonal path a in $\mathbb{R}^2 \setminus P(B)$ so that none of its vertices lie on the knot, none of the vertices of the knot projection lie on a and it crosses the projection only a finite number of times.

Define $a^\# = x_{i_1}^{\epsilon_1} x_{i_2}^{\epsilon_2} \dots x_{i_n}^{\epsilon_n}$ if a crosses $P(A)$ in order of $P(A_{i_1}), P(A_{i_2}), \dots$

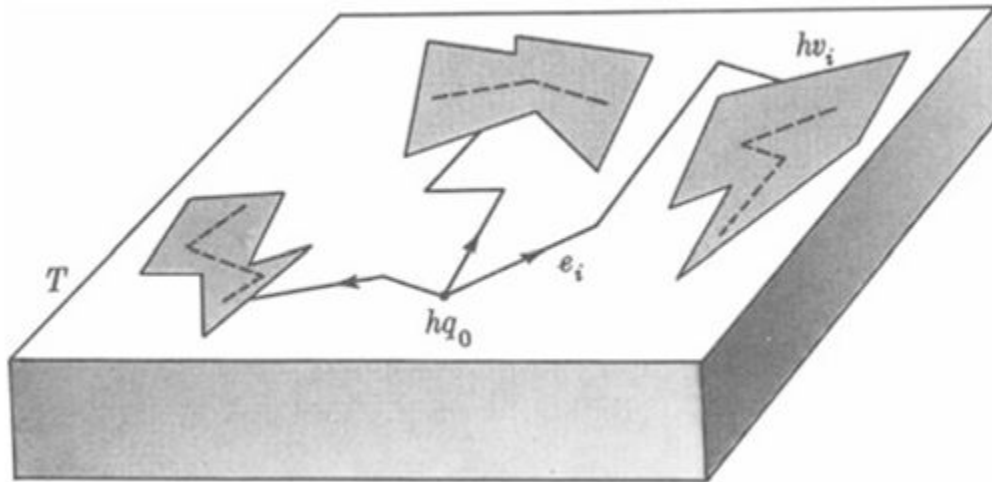
$\epsilon = 1$ if a crosses the knot from left to right $\epsilon = -1$ otherwise. Also define $*a = a(0) \cdot a \cdot (a(1))^{-1}$ Where $\bar{x}$ is a polygonal path which at first comes from p_0 to the point vertically above x on the $z = z_0$ plane along a straight line and comes vertically down to x . Now we define a map ϕ from the free group $F(x)$ where $x = (x_1, x_2, \dots, x_n)$ to $\pi(\mathbb{R}^3 \setminus K', p_0)$.

For any a_i path in $\mathbb{R}^2 \setminus P(B)$ such that $a_i^\# = x_i, \phi(x_i) = *a$

Over Presentation

As $P(B_i)$ are disjoint segmented arcs, we can select n simply connected open sets $V_1, V_2 \dots V_n$ such that $P(B_i) \in V_i$

Let v_i be the simple loop along the boundary of V_i in the counterclockwise direction.

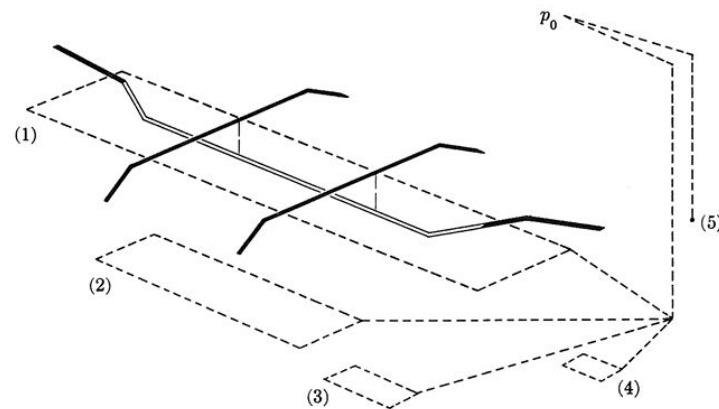


Over Presentation

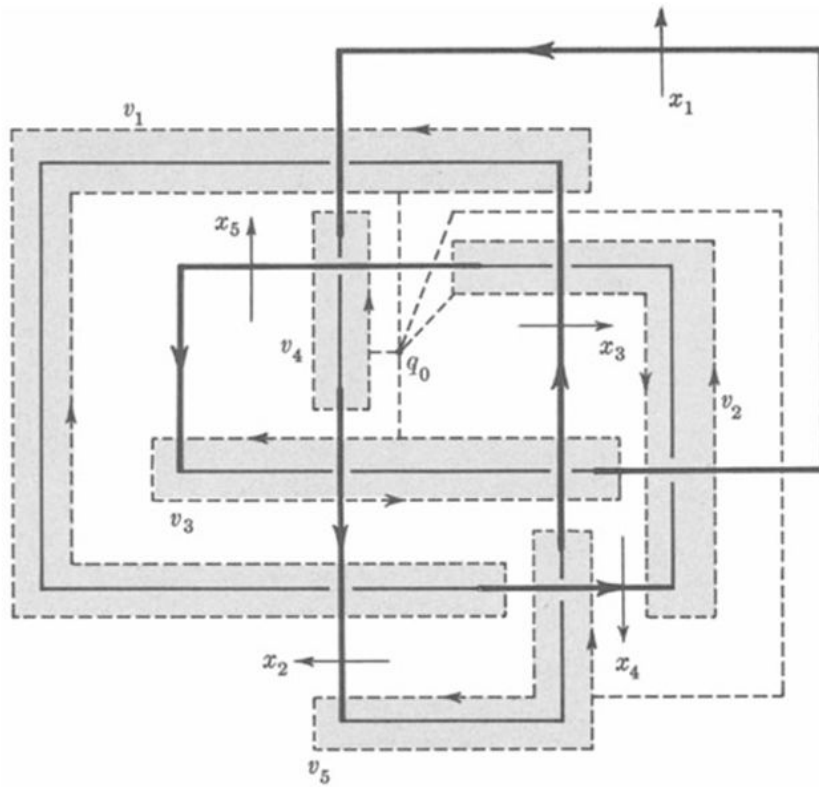
Define n paths, i.e., $c_1, c_2, \dots, c_n$ such that c_i has initial point q_0 and terminal point $v_i(0)$ and no point of c_i belongs to the closure of any V_i . Then the required over presentation of the knot will be

$$G = \langle x_1, x_2, \dots, x_n \mid r_1, r_2, \dots, r_n \rangle_\phi$$

where $r_i = (c_i \cdot v_i \cdot c_i^{-1})^\#$.



Theorem: In an over presentation , any one of the relators is in the consequence of other $n-1$ relators.



Example of over
presentation of a Knot

$$r_1 = x_5 (x_3 x_1 x_2^{-1} x_4^{-1} x_2 x_1^{-1}) x_5^{-1} \quad r_2 = x_3 x_1 x_4 x_1^{-1} x_3^{-1} x_5^{-1}$$

$$r_3 = x_2 x_5 x_2^{-1} x_3 x_1^{-1} x_3^{-1} \quad r_4 = x_5 x_1 x_5^{-1} x_2^{-1}$$

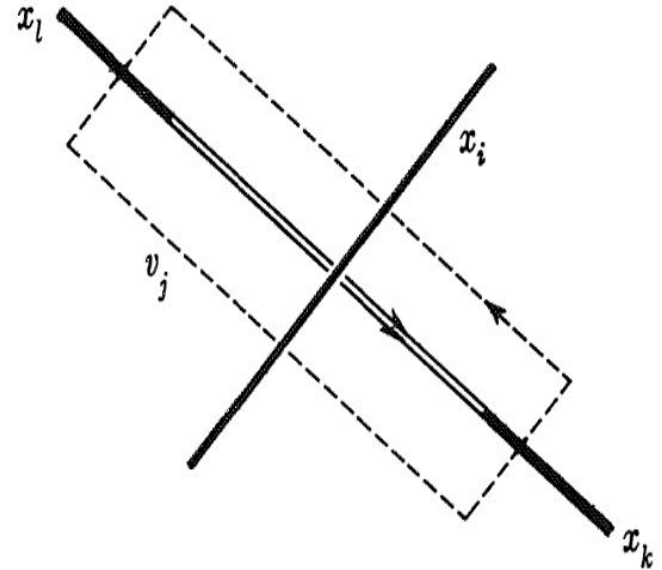
$$r_5 = x_5 x_3 x_1 (x_4^{-1} x_3^{-1} x_4 x_2) x_1^{-1} x_3^{-1} x_5^{-1}$$

Wirtinger Presentation

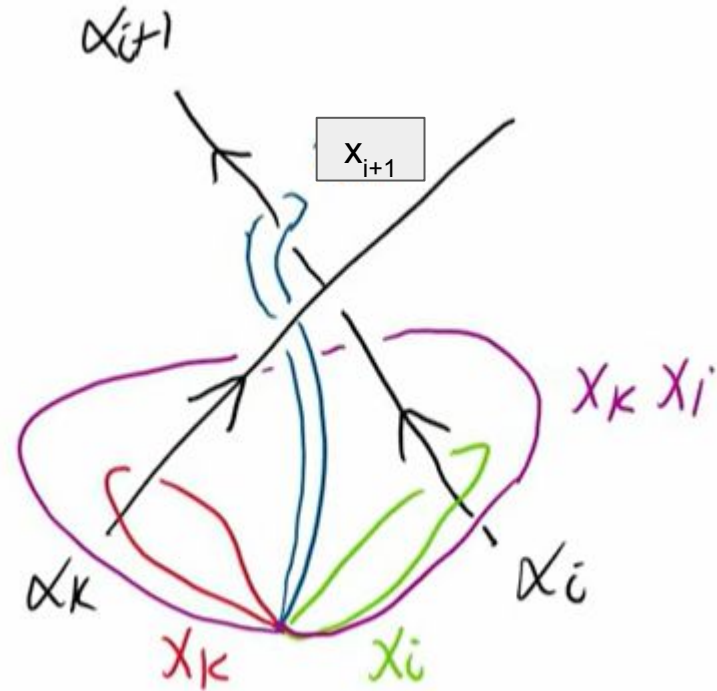
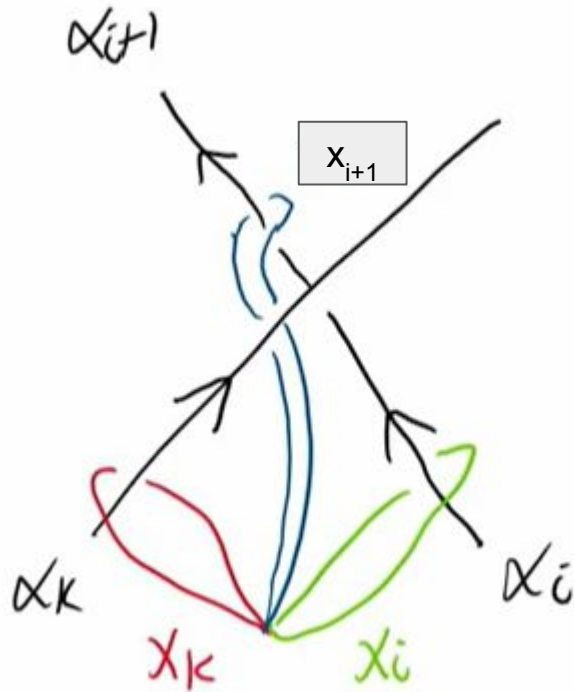
Invoking the Tietze operations I and I' n times each, we can modify the over presentation into

$$G = \langle x_1, x_2, \dots, x_n \mid v_1^\#, v_2^\#, \dots, v_n^\# \rangle_\phi$$

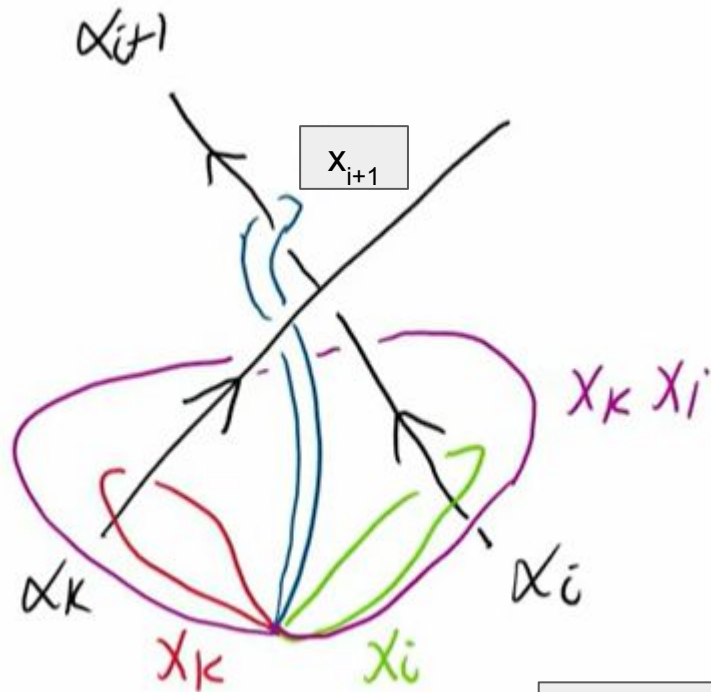
If each underpass contains exactly one undercrossing, and each path v_i cuts the projected overpasses in just four places, then this projection is called a Wirtinger projection. Wirtinger presentation is favoured over other presentations because all the relators are of the type $x_k = x_i^\epsilon x_l x_i^{-\epsilon}$, $\epsilon = \pm 1$



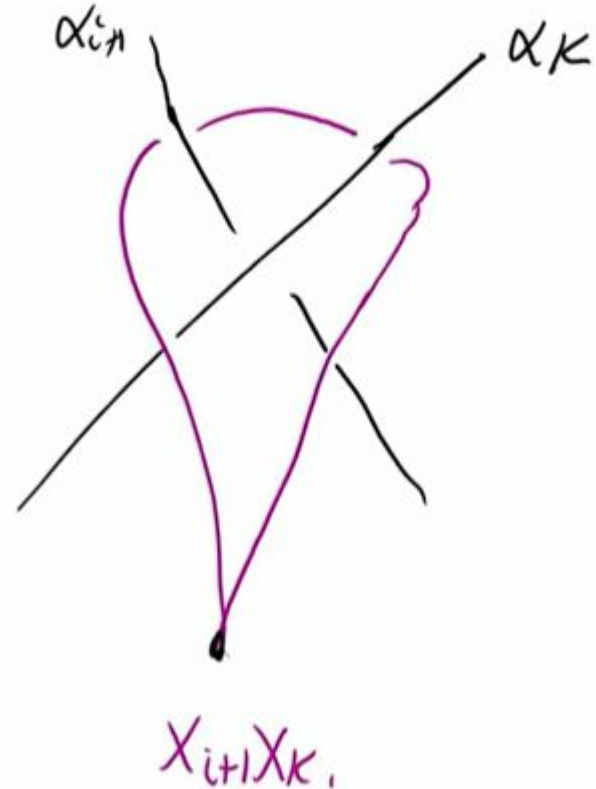
Wirtinger Presentation: Principle



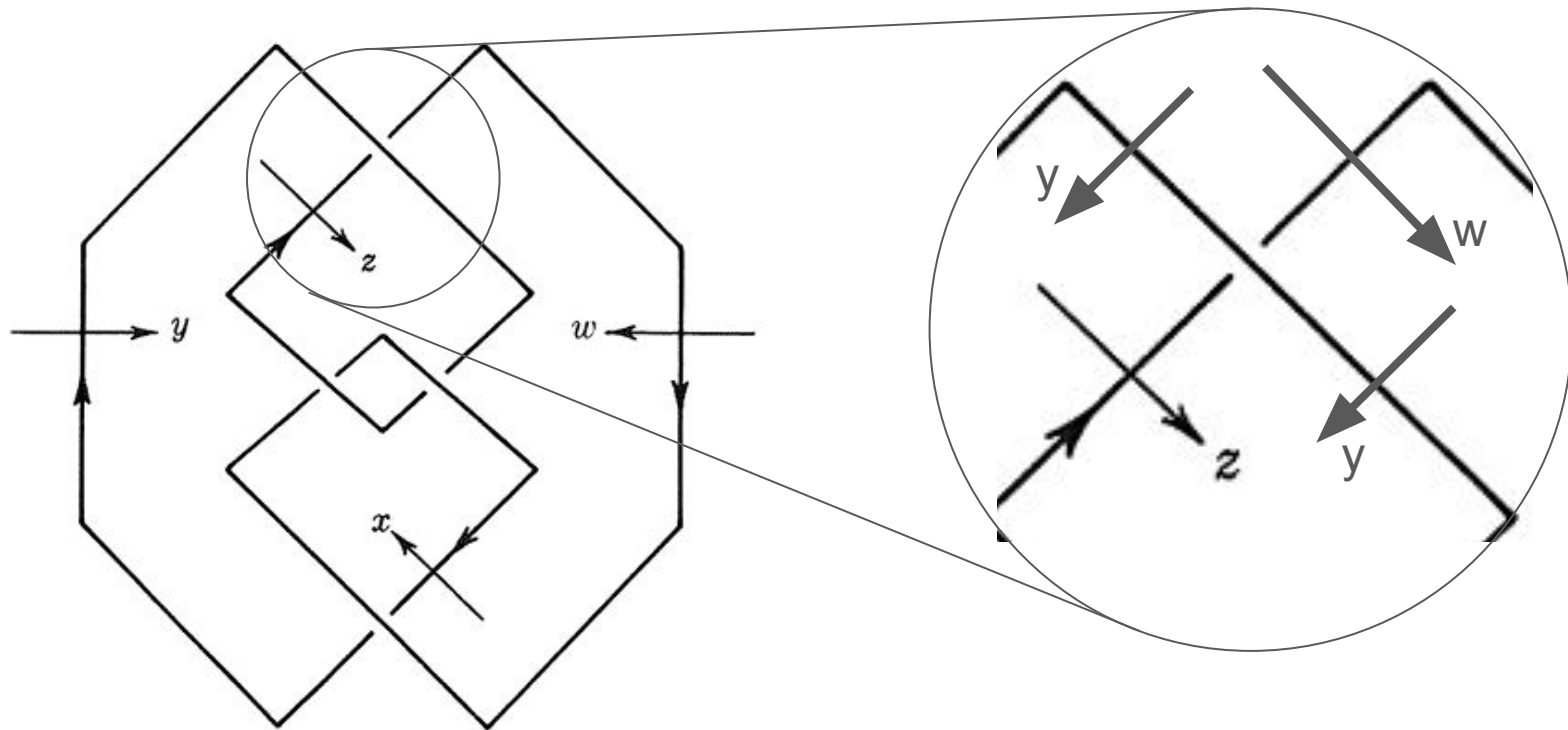
Wirtinger Presentation: Principle



Thus $x_k x_i = x_{i+1} x_k$



Wirtinger presentation of figure-8 Knot



Wirtinger presentation of figure-8 Knot

$$G = \langle x, y, z, w \mid x^{-1}z^{-1}wz, y^{-1}wxw^{-1}, z^{-1}x^{-1}yx \rangle$$



Replace z with $x^{-1}yx$



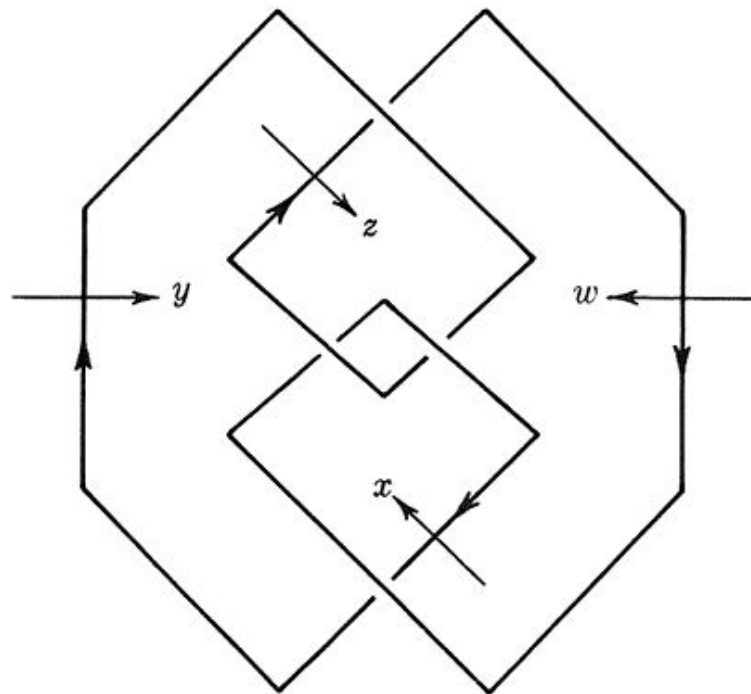
$$G = \langle x, y, w \mid x^{-1}y^{-1}wx^{-1}y = 1, y = wxw^{-1} \rangle$$



Replace w with $x^{-1}yxy^{-1}x$



$$G = \langle x, y \mid yx^{-1}yxy^{-1} = x^{-1}yxy^{-1}x \rangle$$



Exploration: Knot Group of Trefoil

Activity

References

- [1] Introduction to Knot Theory (1963), Crowell, Fox
- [2] Solvable and Unsolvable Problems (1954), Alan Turing
- [3] A note on lattice knots, Sasha Anan'in, Alexandre Grishkov, Dmitrii Korshunov
- [4] The algebraic topology playlist, Gabriel Islambouli
[\[https://www.youtube.com/playlist?list=PLevbjS885INS9Nk8ve13FE5tMdZejsTrE\]](https://www.youtube.com/playlist?list=PLevbjS885INS9Nk8ve13FE5tMdZejsTrE)
- [5] The fundamental group of the torus is abelian, 2006
[\[https://www.youtube.com/watch?v=nLcr-DWVEto\]](https://www.youtube.com/watch?v=nLcr-DWVEto)